Behind the toxic cocktail: Toxicokinetic mechanisms of pesticide mixture interactions in earthworms

Supplementary information

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1 Earthworm culture

1.1 Breeding and experiment soil

The soil used in the culture of earthworms is collected from the top 10 cm of a meadow in Versailles called "Les Closeaux". It is then dried before being milled at < 2 mm.

Table 1.1: Main characteristics of the breeding soil from Versailles. Table from Bart et al. (2017).

Characteristics	Value
Clay (< 2 µm, g/kg)	226.0
Fine silt (2-20 µm, g/kg)	174.1
Coarse silt (20-50 µm, g/kg)	298.9
Fine sand (50-200 µm, g/kg)	239.1
Coarse sand (200-2000 µm, g/kg)	47.9
$CaCO_3$ total (g/kg)	23.3
Organic matter (g/kg)	32.6
P_2O_5 (g/kg)	0.1
Organic carbon (g/kg)	18.9
Total nitrogen (N) (g/kg)	1.5
C/N	12.5
pH	7.5
Total Cu (mg/kg)	25.2

1.2 DNA barcoding

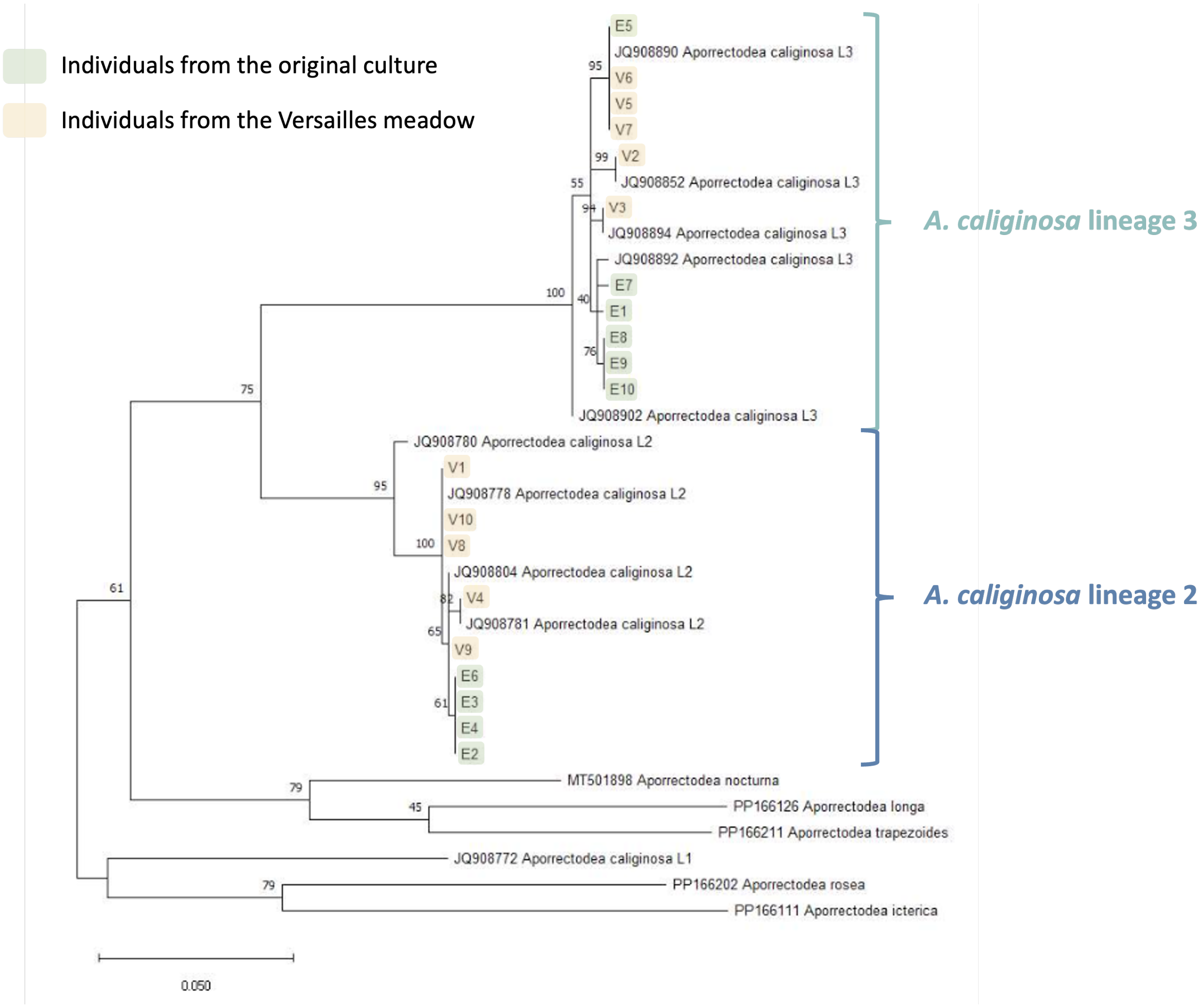

Figure 1.1: Results of the DNA barcoding of 10 individuals from the culture (highlighted in yellow)

The phylogenetic tree was realised thanks to maximum likelyhood bootstrap method (1000 simulations) with the Tamura-Nei model (nucleotides). Bootstrap frequency is indicated on each branch.

Aporrectodea caliginosa is a cryptic species complex comprising three lineages. Our earthworm culture is known to include individuals from two closely related lineages (2 and 3). While these lineages could potentially exhibit differences in sensitivity, previous research on *Gammarus roeselii* has shown that tolerance within cryptic species complexes is not always lineage dependent (Kabus et al., 2024).

2 Experiments

2.1 Single substances experiments

2.1.1 Toxicokinetic experiments (TK & TK BIS experiments)

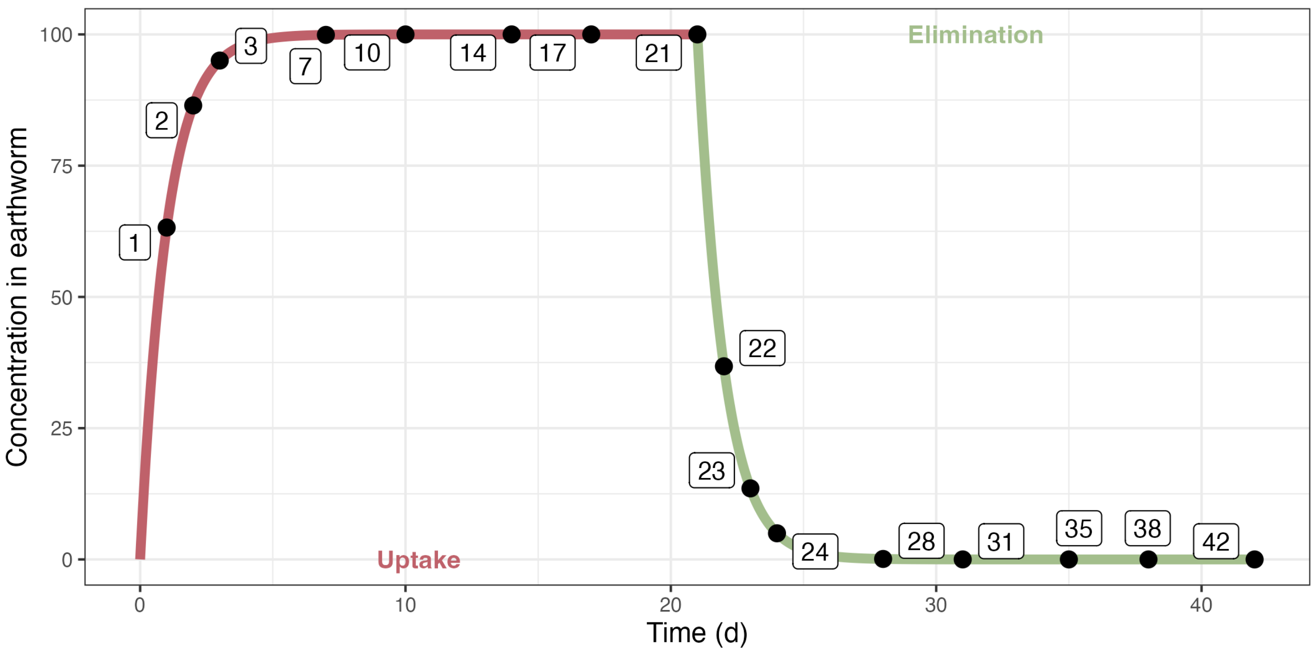

Figure 2.1: Design of the initial toxicokinetic experiments

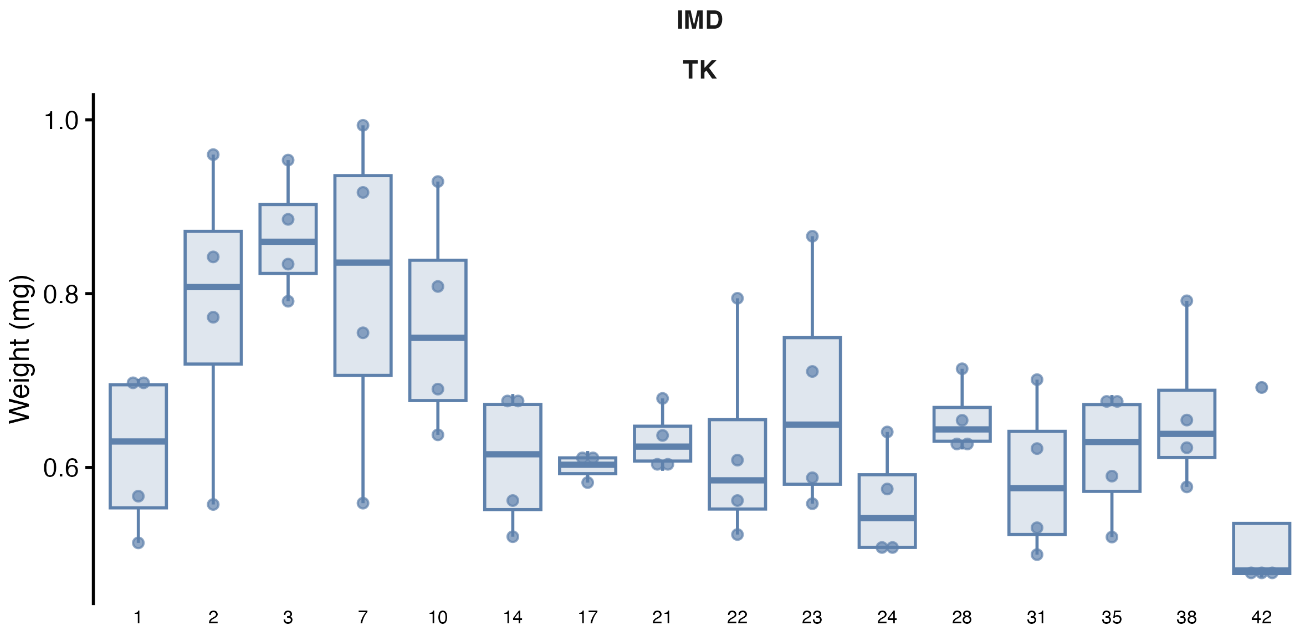

Figure 2.2: Initial weights (with gut content) of earthworms for the IMD experiment.

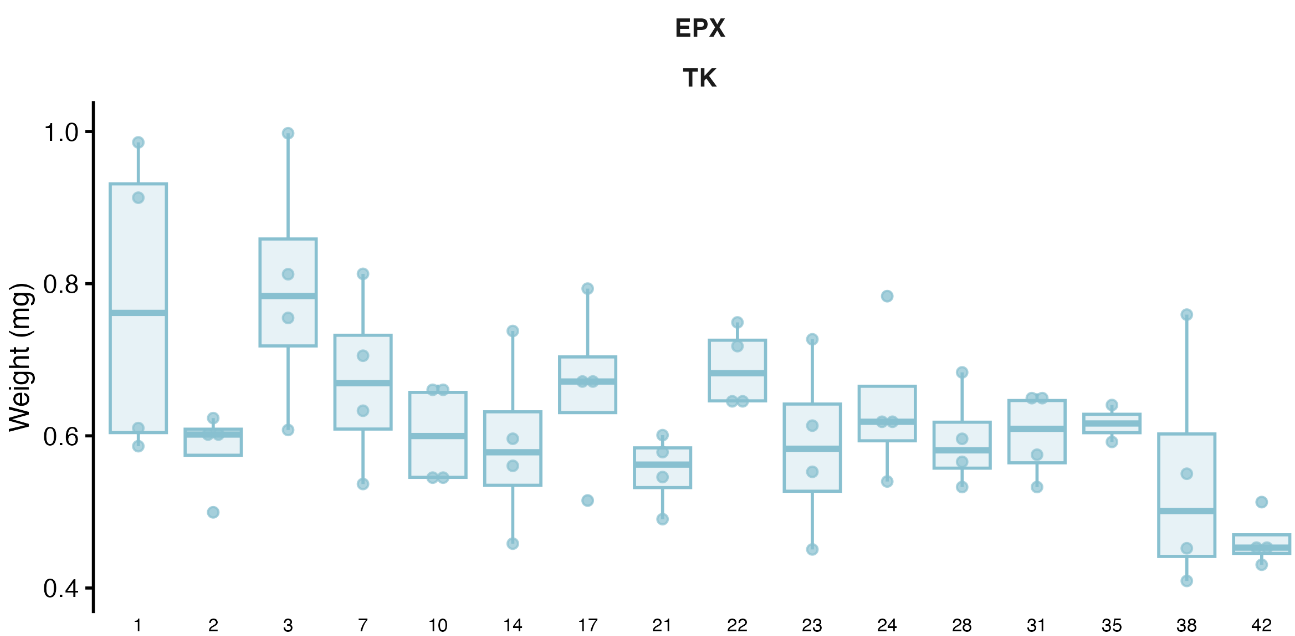

Figure 2.3: Initial weights (with gut content) of earthworms for the EPX experiment.

Table 2.1: Initial weight of earthworms (with gut content, mg) in the toxicokinetic experiments.

Molecule	Min	1st Q	Mean	Median	3rd Q	Max	n
IMD (TK)	474.0	565.7	662.5	635.1	711.3	993.7	64
IMD (TK BIS)	350.0	373.2	402.8	408.8	416.5	490.8	9

Molecule	Min	1st Q	Mean	Median	3rd Q	Max	n
EPX (TK)	409.3	544.7	622.8	605.8	672.6	997.7	62
EPX (TK BIS)	386.7	408.0	462.9	414.4	467.4	736.7	9

2.1.2 Link between weight with and without gut content

Since experimental data include weights with and without gut content, we established a quantitative relationship between them to allow conversion and ensure consistency across datasets. The relationship is made with the data from the first toxicokinetic experiments with the weight of the earthworm before and after the 24h in Petri dishes (and massage if needed). We make the assumption that the variation in body weight without gut content is negligeable during the 24 hours. The results are shown Figure 2.4.

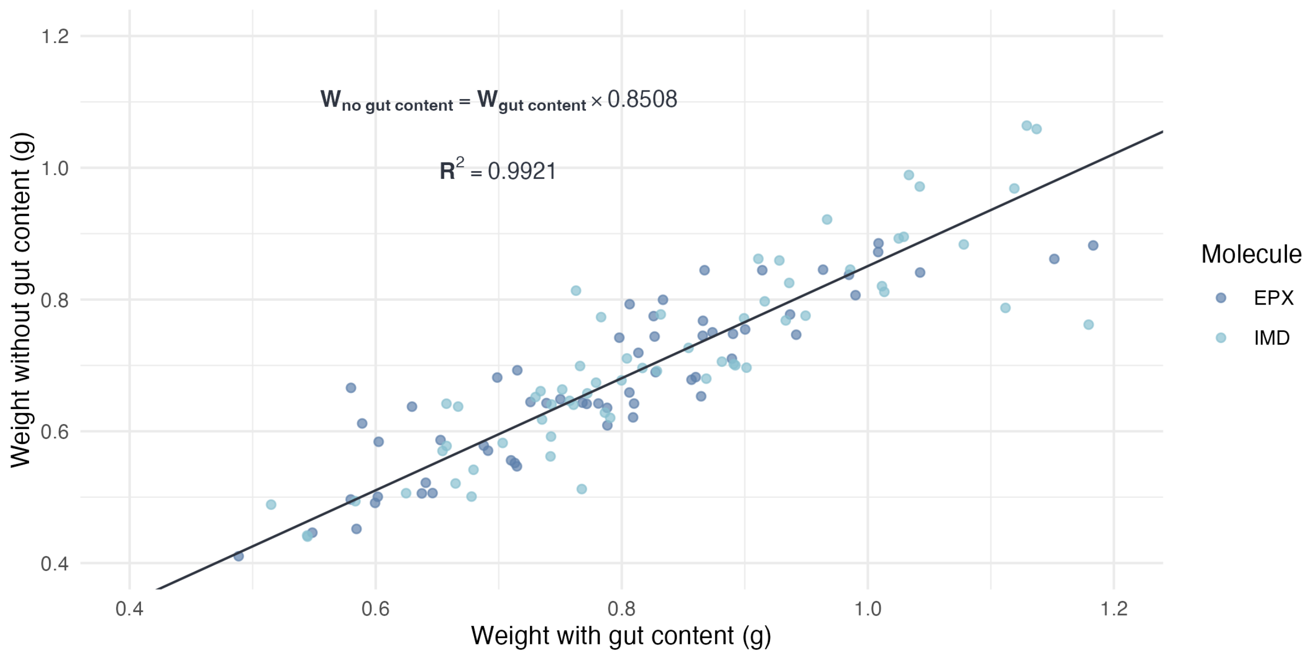

Figure 2.4: Relationship between weight with and without gut content.

2.1.3 Depuration kinetic in Pétri dishes (GUT experiment)

Depuration of the earthworm gut content was modeled as a first-order decay (Equation 2.1), informed by a small supporting experiment: ten earthworms were kept individually in 200 g of Closeaux soil with 3 g of horse dung for one week, then transferred to Petri dishes lined with moist filter paper and weighed at regular intervals. After 32 hours, the worms were gently massaged to void the remaining gut contents and weighed again, providing data to estimate the gut-clearance rate.

$$\frac{dP_{Weight}}{dt} = -\frac{\ln 2}{DT_{50,\text{gut}}} \left(P_{Weight} - P^{\infty}_{Weight}\right) \tag{2.1}$$

$$P_{Weight} = P_{Weight}^{\infty} + (100 - P_{Weight}^{\infty}) \, \exp\left(-\frac{\ln 2}{DT_{50,\text{gut}}} \times t\right)$$

With :

- t : Time (h)
- P_{Weight} : Percentage of initial weight (%, 0-100)
- P_{Weight}^{∞} : Percentage of initial weight when gut is fully emptied (%)
- $DT_{50,\text{gut}}$: Half-life of the gut content (h)

For the model, we consider that weights measured after massage are weights measured after a significant amount of time has passed. Therefore, the corresponding percentages of initial weight inform P_{Weight}^{∞}.

The estimated value for $DT_{50,\text{gut}}$ and P_{Weight}^{∞} are 4.37 [3.11 ; 7.32] hours and 86.2 [84.6 ; 87.9] %, respectively.

P_{Weight}^{∞} estimated here is consistent with the one estimated in Section 2.1.2.

2.2 Mixture experiment

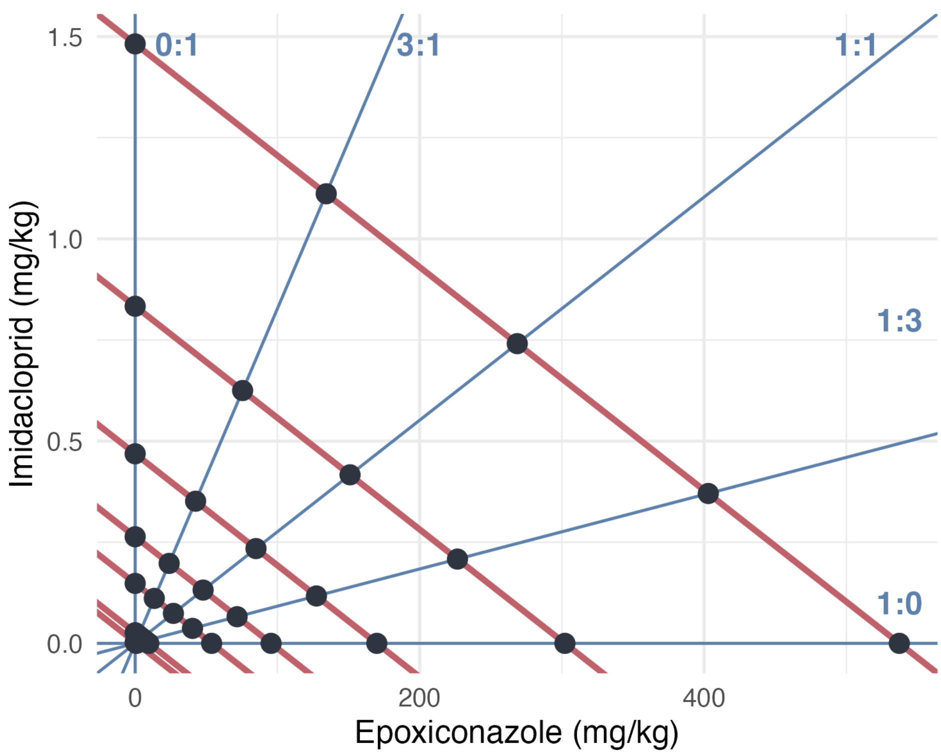

Figure 2.5: Experimental design of the mixture experiment (MIX). Blue lines represent the ratio if TU tested and the red lines correspond to effect isoboles under concentration addition hypothesis.

Ratio are labeled with letters :

- E : 1:0 (EPX only)
- F : 3:1
- G : 1:1
- H : 1:3
- I : 0:1 (IMD only)

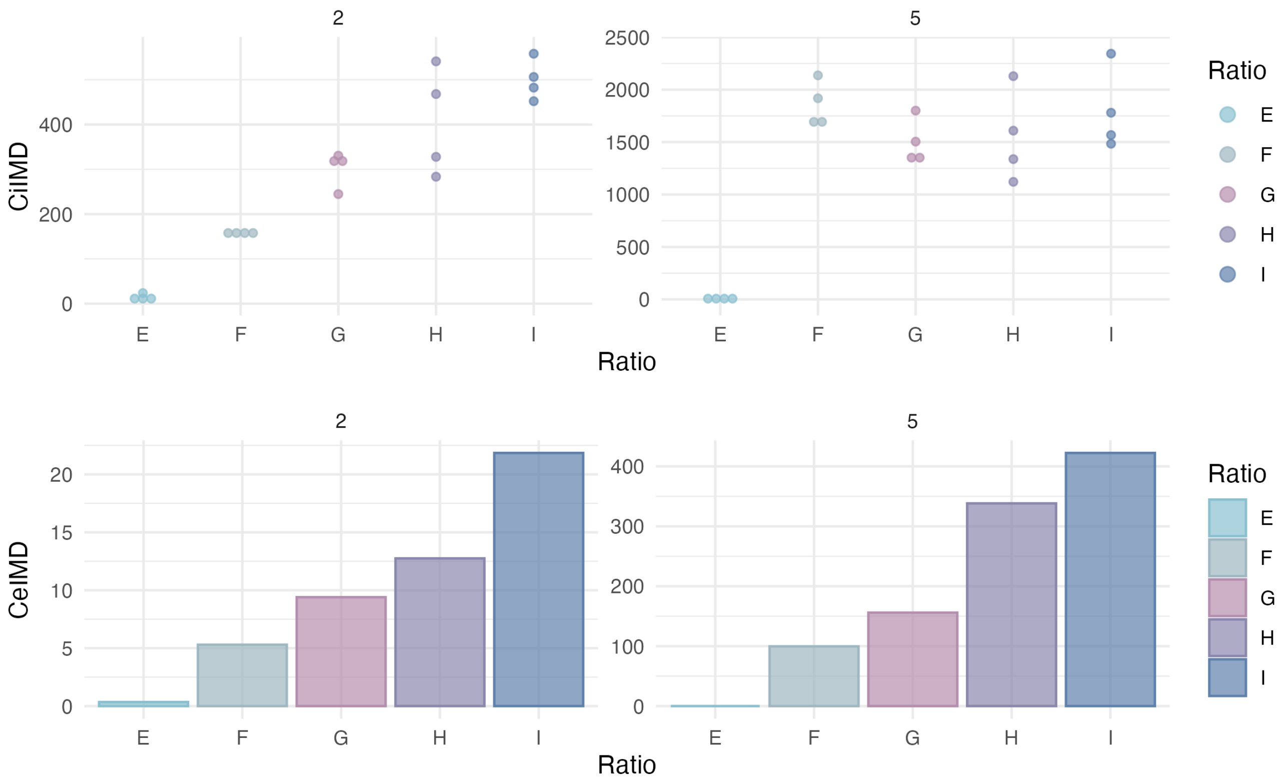

Figure 2.6: Results of the chemical analysis of internal concentrations in IMD of earthworms exposed to the mixture.

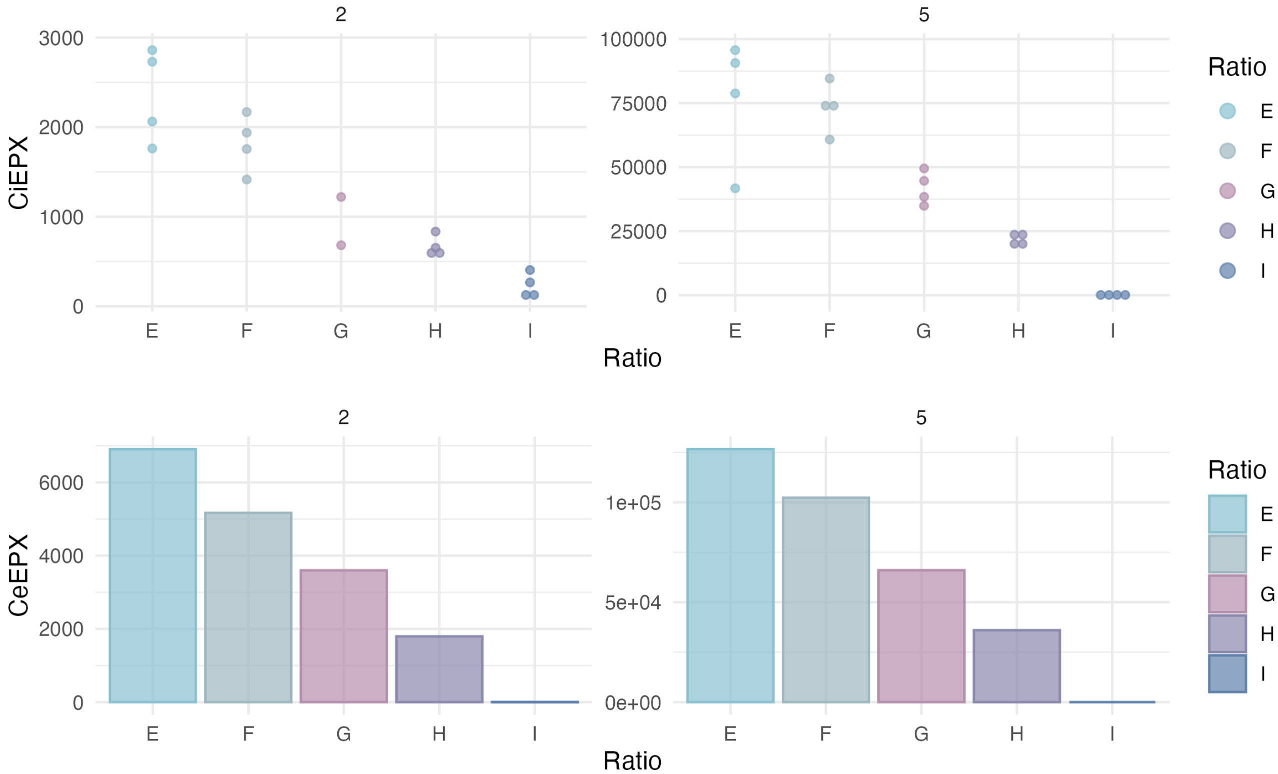

Figure 2.7: Results of the chemical analysis of internal concentrations in EPX of earthworms exposed to the mixture.

3 Toxicokinetics modeling - Single substances

3.1 Priors used

Priors are the same between models :

- Distrib(kuIMD, LogUniform, 0.0001, 100000); # min, max
- Distrib(keIMD, LogUniform, 0.0001, 100000); # min, max
- Distrib(kperiph, LogUniform, 0.00001, 100000); # min, max
- Distrib(phi, Uniform, 0.0, 1); # min, max
- Distrib(a_growth, TruncNormal, 0.005, 0.0010, -0.5, 1); # mean, sd, min, max
- Distrib (Vr_a_growth, TruncNormal, 0.0015, 0.01, 0, 1); # mean, sd, min, max
- Distrib(Sigma_W, TruncNormal, 0.03, 0.01, 0.0001, 1); # mean, sd, min, max
- Distrib(Sigma_CiIMD, TruncNormal, 3.0, 3.0, 1, 1000); # mean, sd, min, max

Log-Likelihood calculations :

- Likelihood(CiIMD, LogNormal, Prediction(CiIMD), Sigma_CiIMD);
- Likelihood(Weight, Normal, Prediction(Weight), Sigma_W);

3.2 Imidacloprid

3.2.1 Model C1

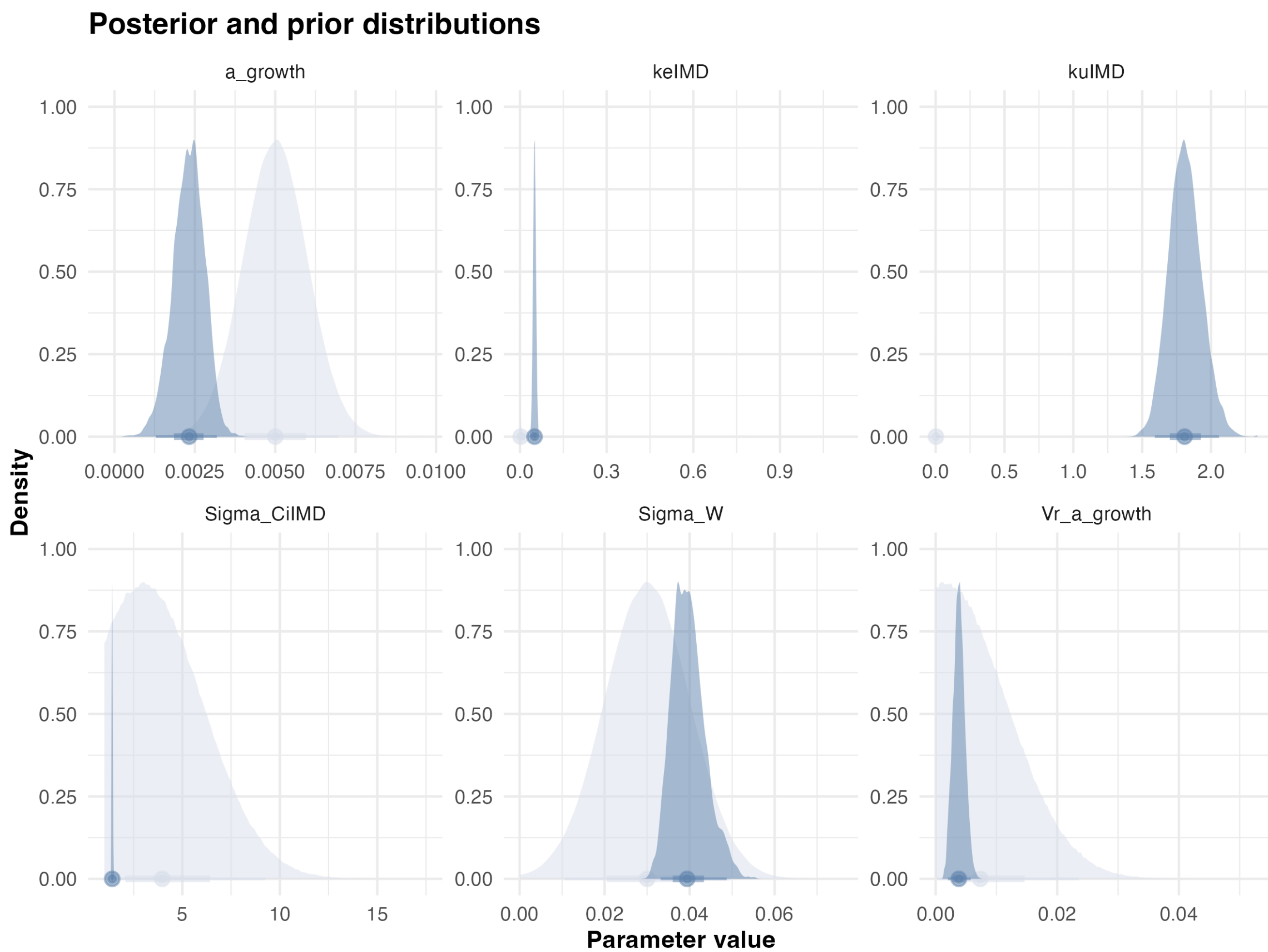

Figure 3.1: Posteriors.

Statistic	k_u	k_e	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	2.0	0.050	0.0031	0.0017	0.045	1.4	36
Q2.5%	1.6	0.041	0.0013	0.0020	0.033	1.4	32
Q97.5%	2.1	0.060	0.0032	0.0058	0.049	1.5	41

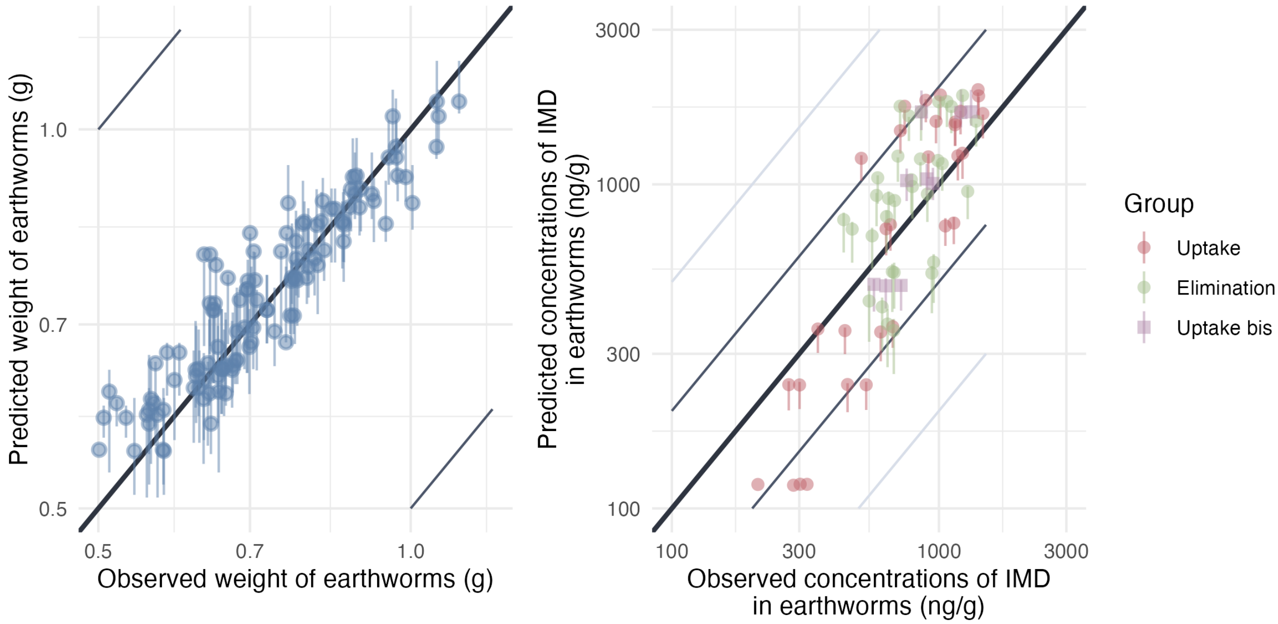

Figure 3.2: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.2.2 Model C1P

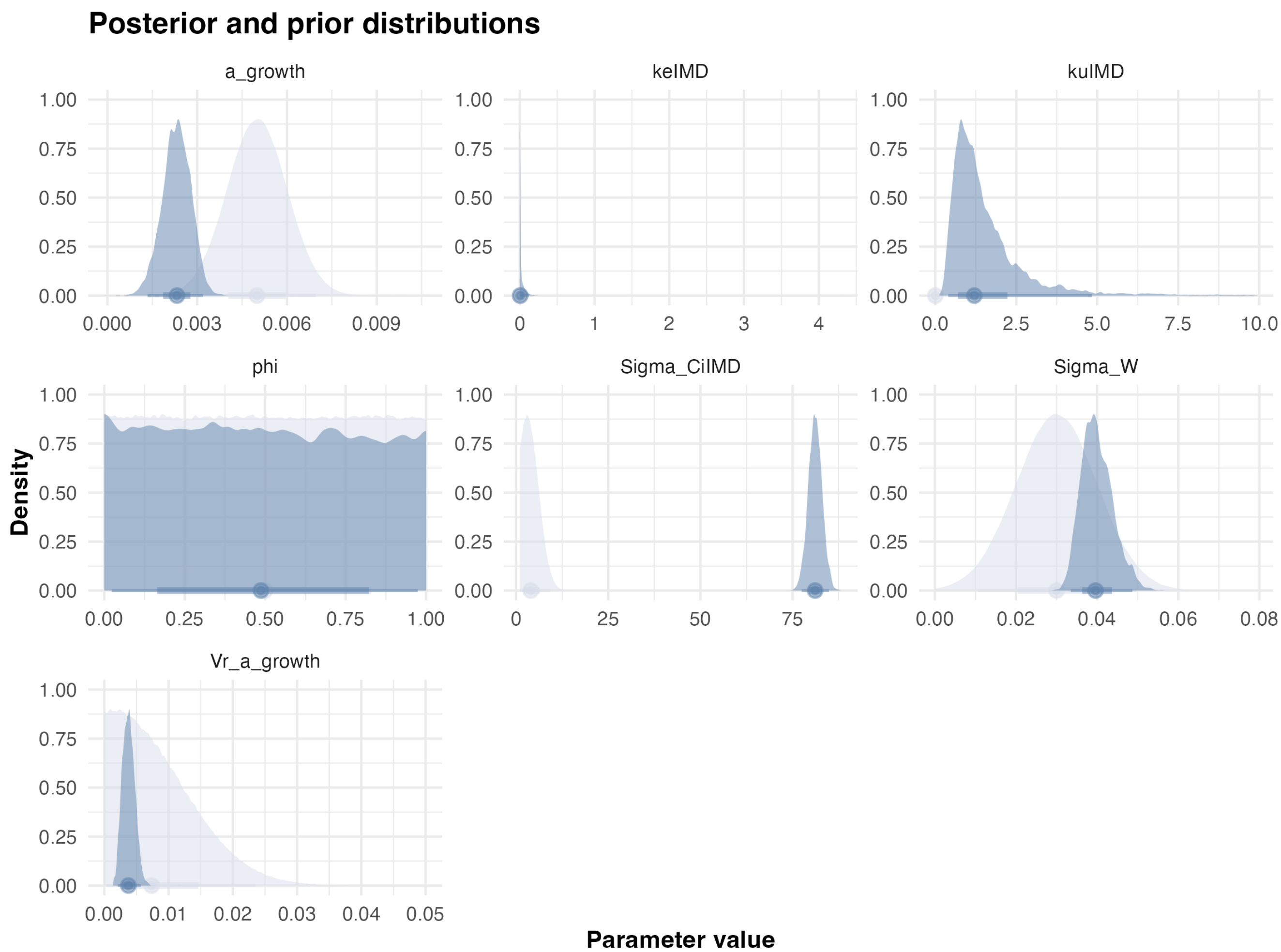

Figure 3.3: Posteriors.

Statistic	k_u	k_e	ϕ	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	0.68	0.00014	0.0097	0.0023	0.0029	0.039	80	3200
Q2.5%	0.4	0.00012	0.023	0.0013	0.0021	0.034	77	17
Q97.5%	5.1	0.12	0.97	0.0032	0.0057	0.049	85	10000

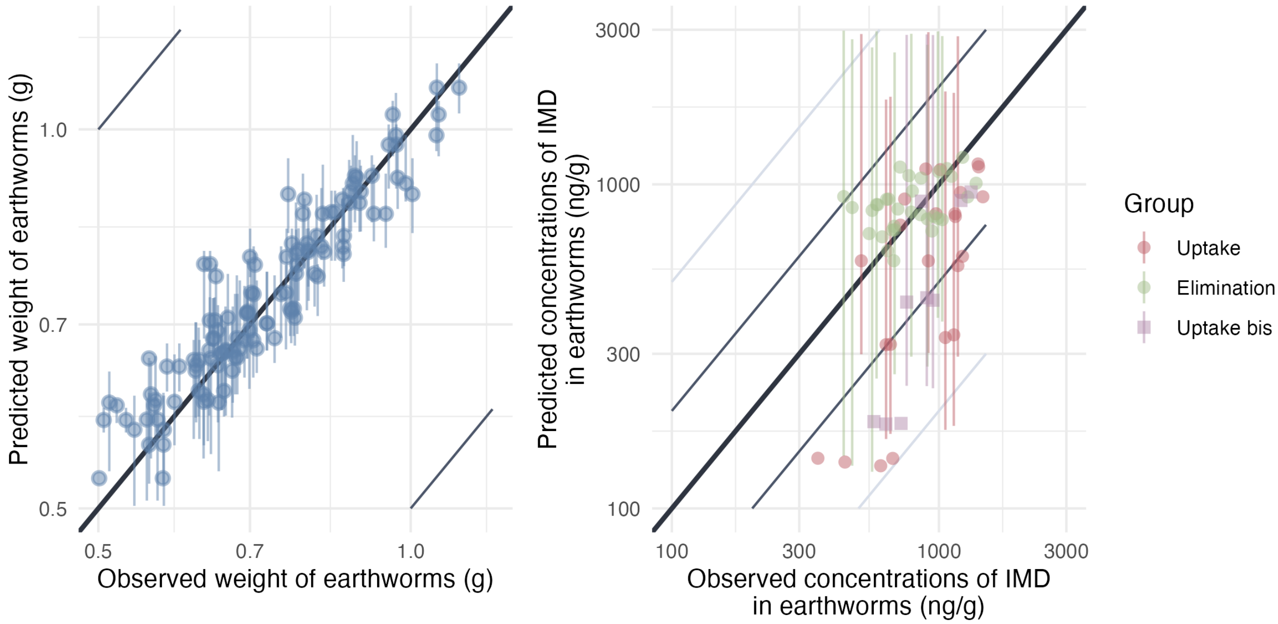

Figure 3.4: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.2.3 Model C2N

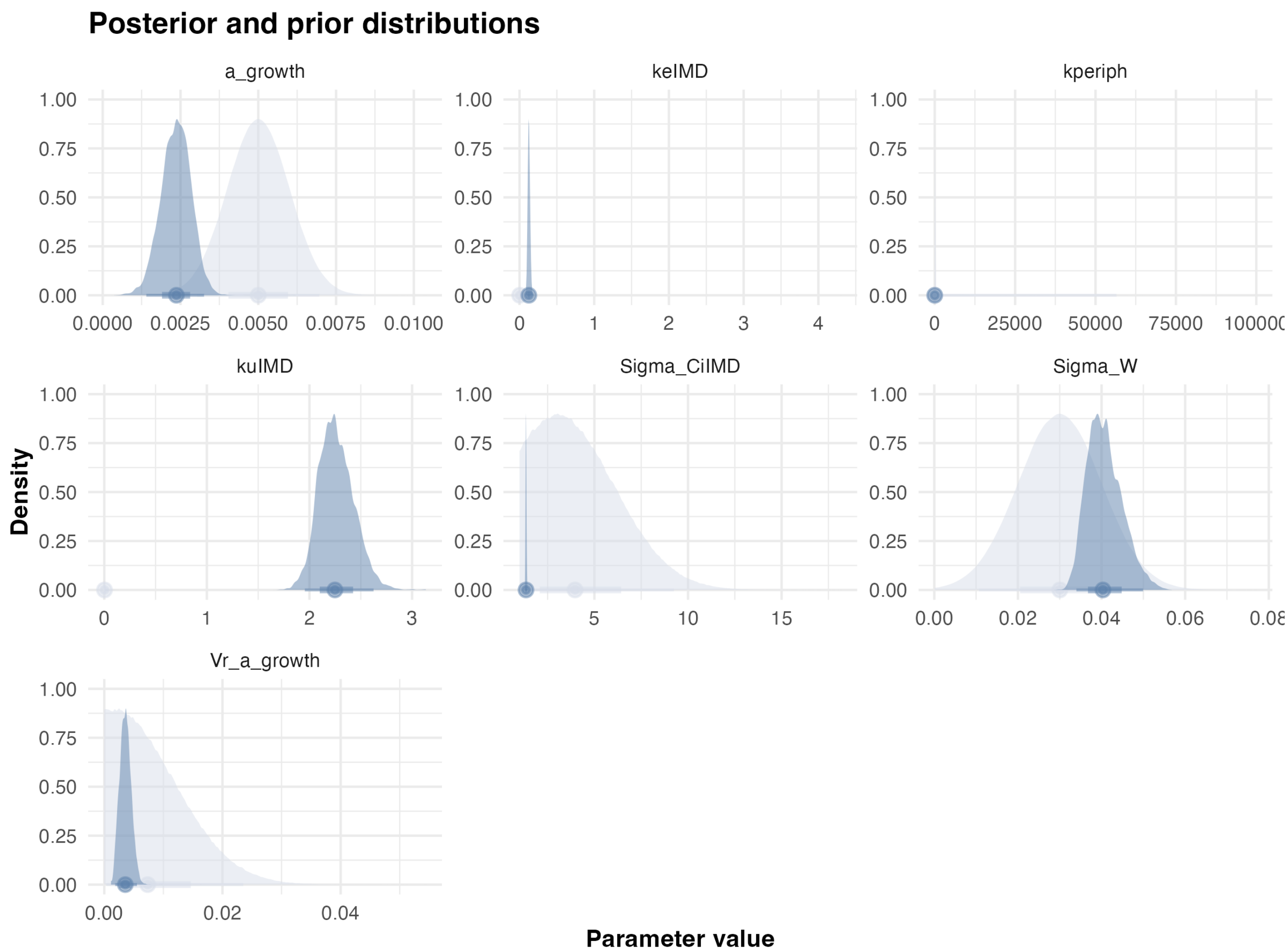

Figure 3.5: Posteriors.

Statistic	k_u	k_e	k_{periph}	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	2.2	0.13	0.058	0.0033	0.0018	0.046	1.3	18
Q2.5%	2.0	0.010	0.030	0.0014	0.0018	0.034	1.3	16
Q97.5%	2.6	0.16	0.073	0.0033	0.0055	0.050	1.4	21

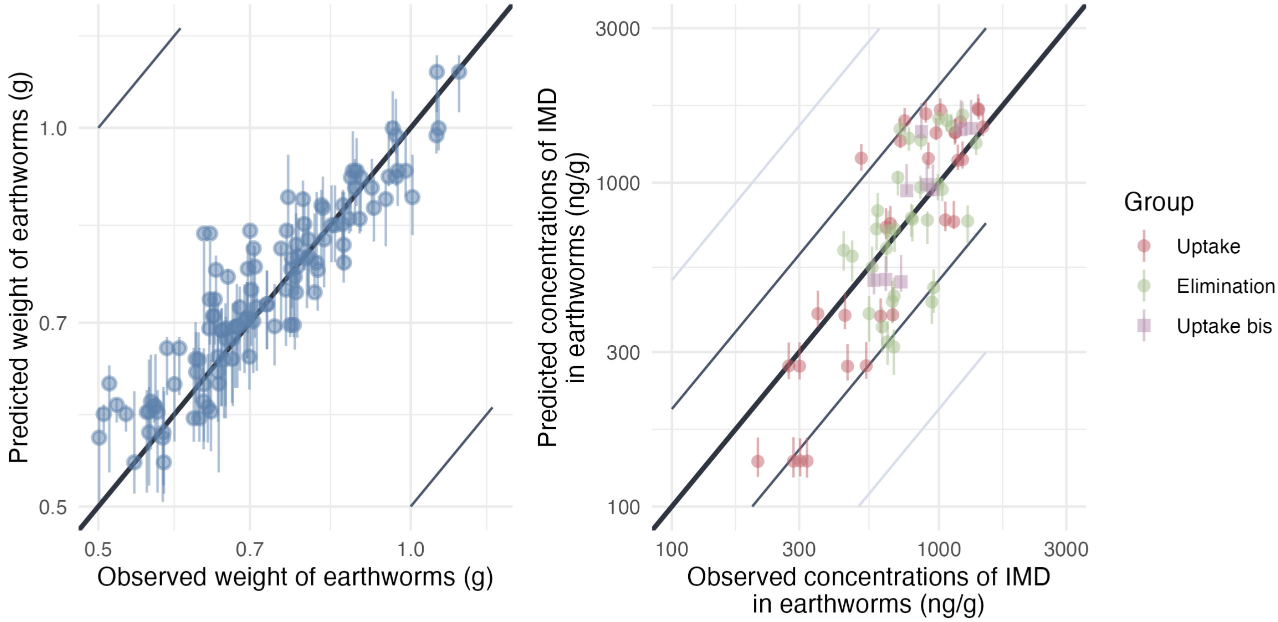

Figure 3.6: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.2.4 Model C2

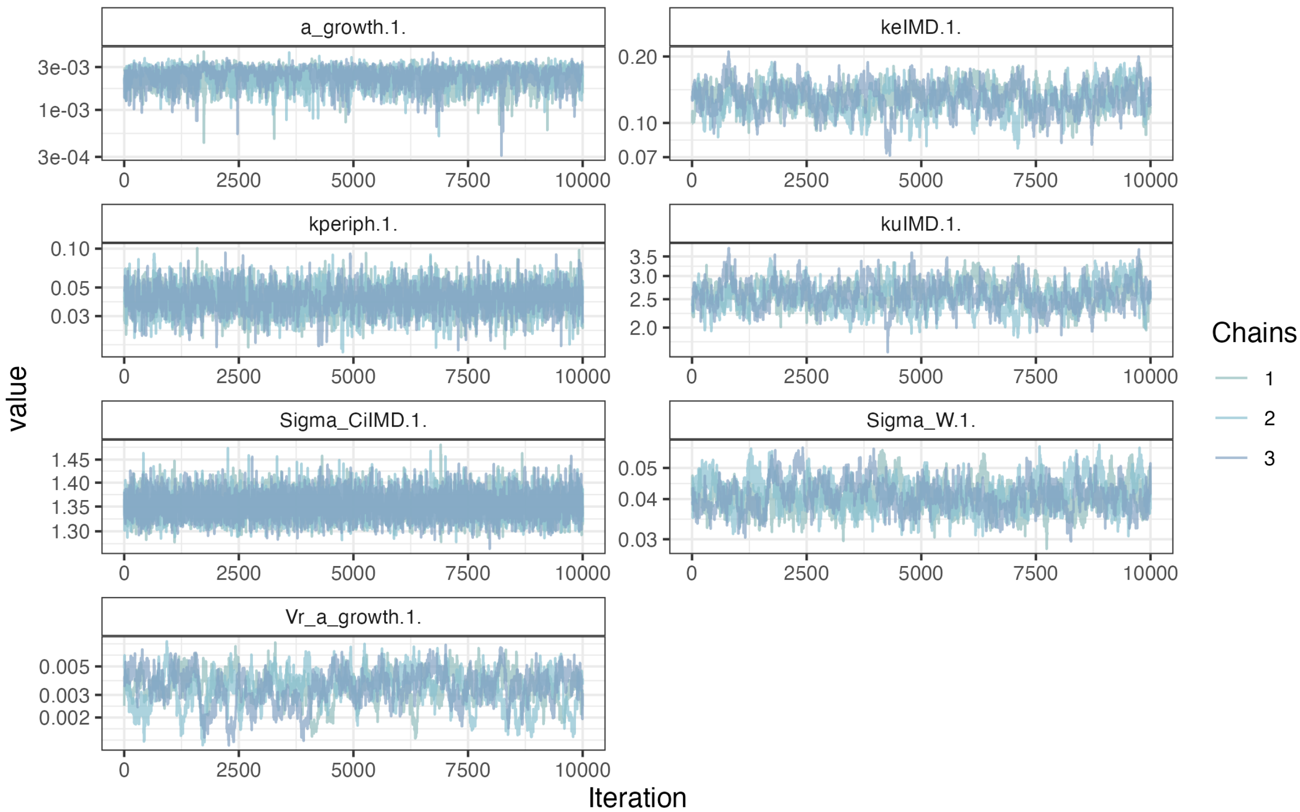

Figure 3.7: MCMC chains of the model.

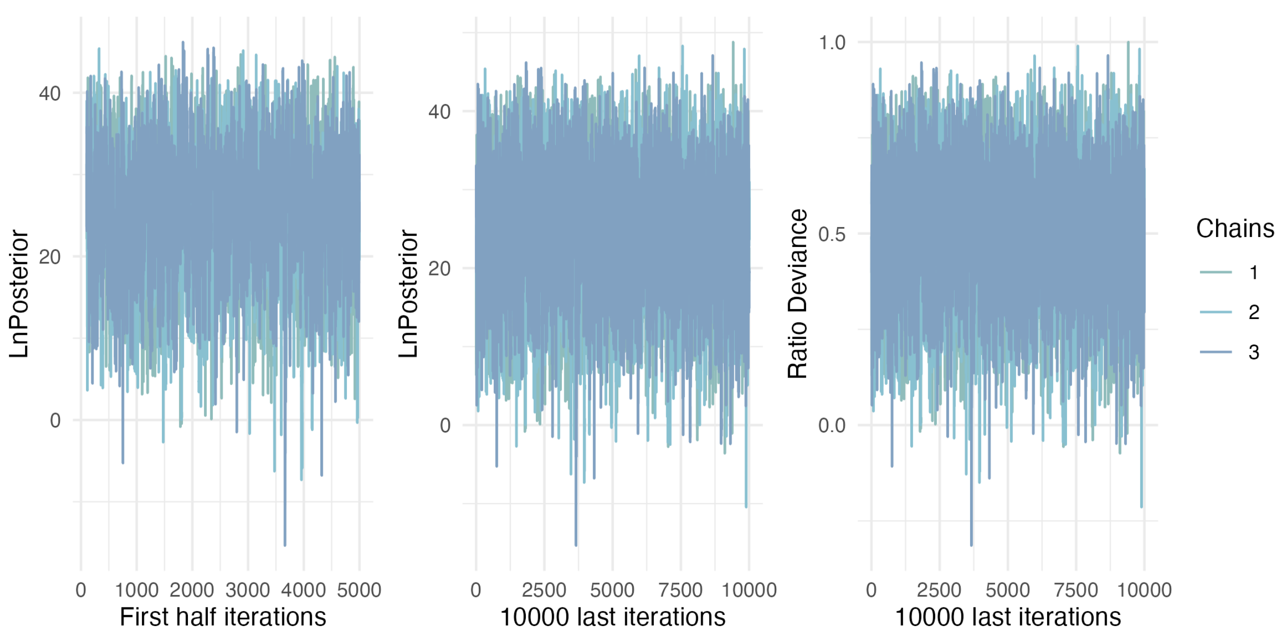

Figure 3.8: Likelihood of chains.

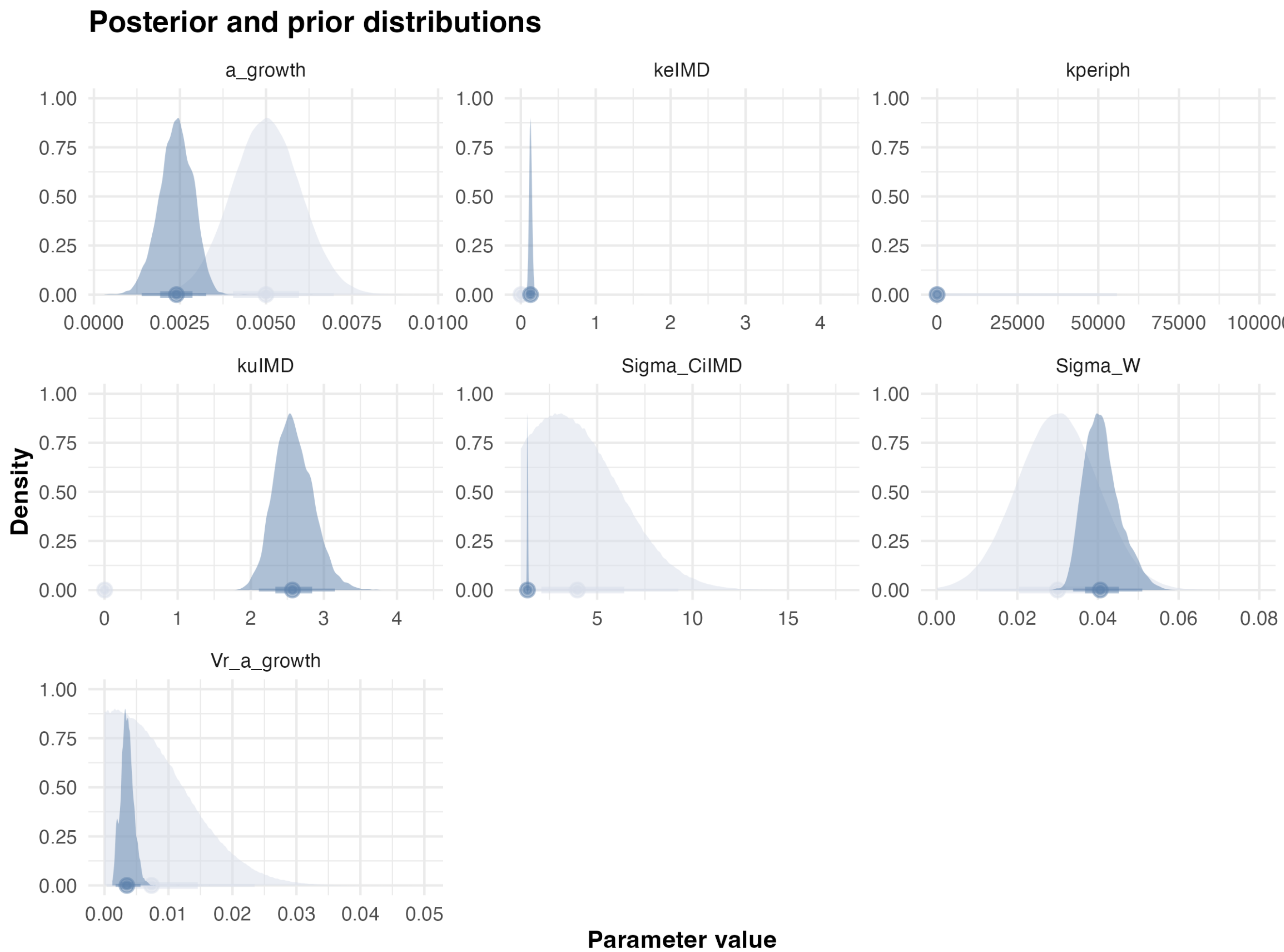

Figure 3.9: Posteriors.

Statistic	k_u	k_e	k_{periph}	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	2.4	0.12	0.045	0.0026	0.0030	0.039	1.4	21
Q2.5%	2.1	0.096	0.025	0.0014	0.0017	0.034	1.3	18
Q97.5%	3.2	0.17	0.063	0.0033	0.0056	0.051	1.4	23

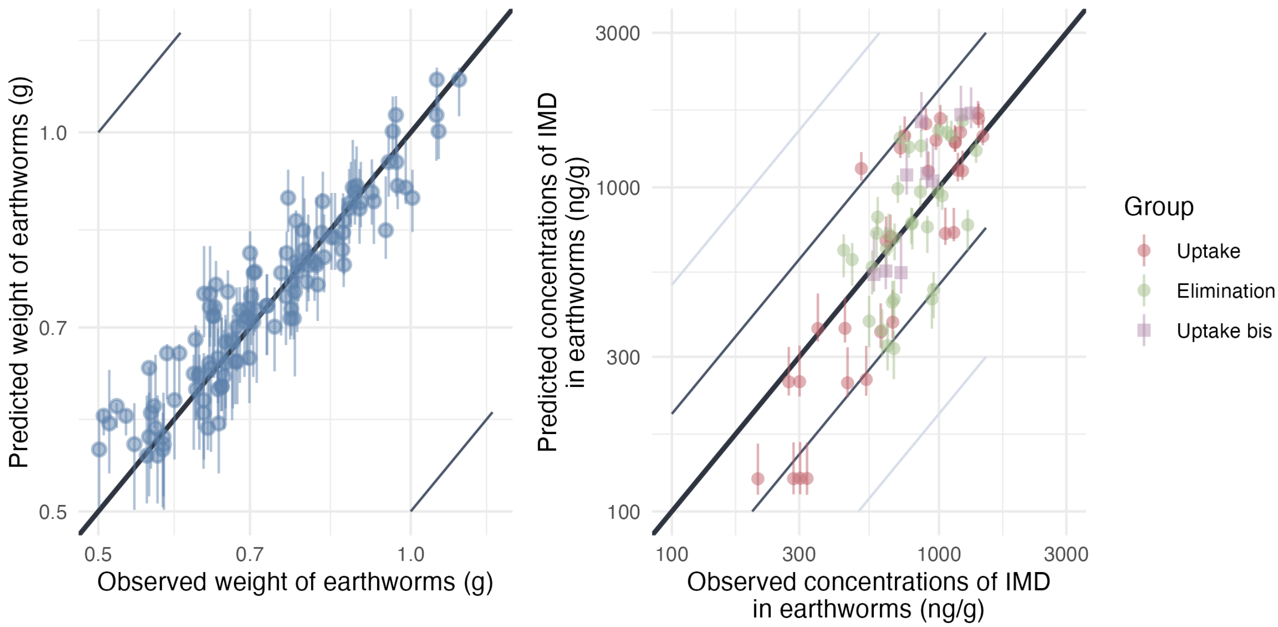

Figure 3.10: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

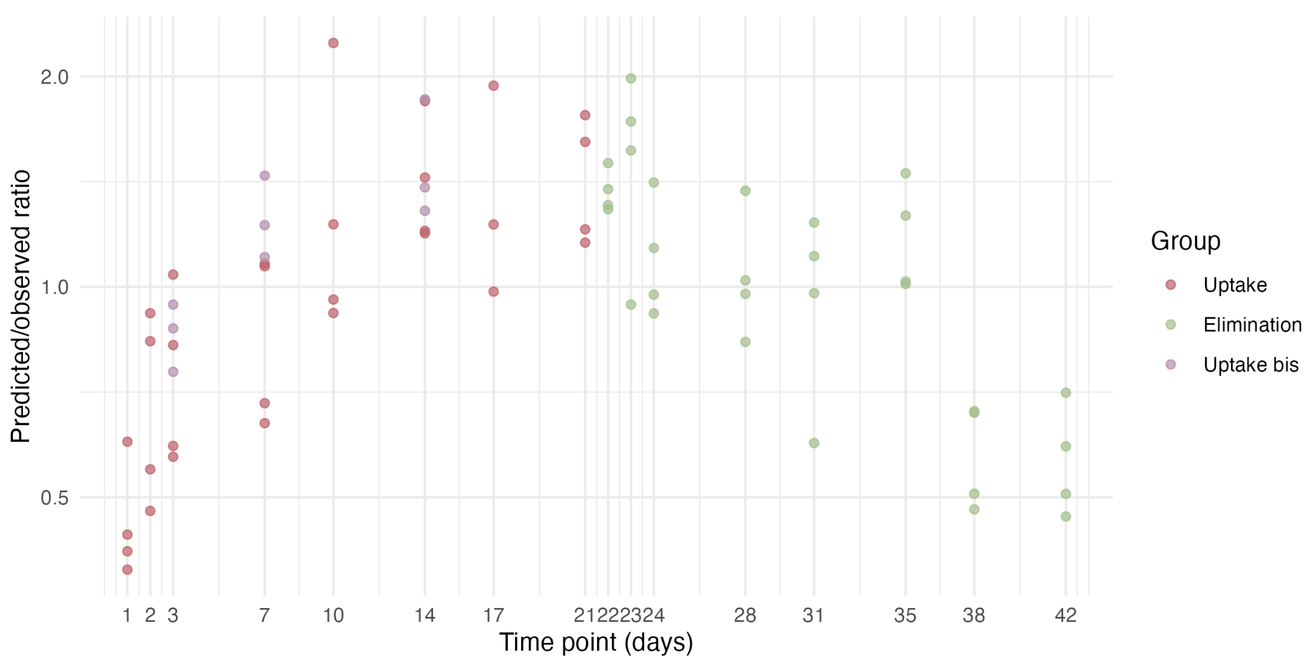

Figure 3.11: Model predicted/observed ratios.

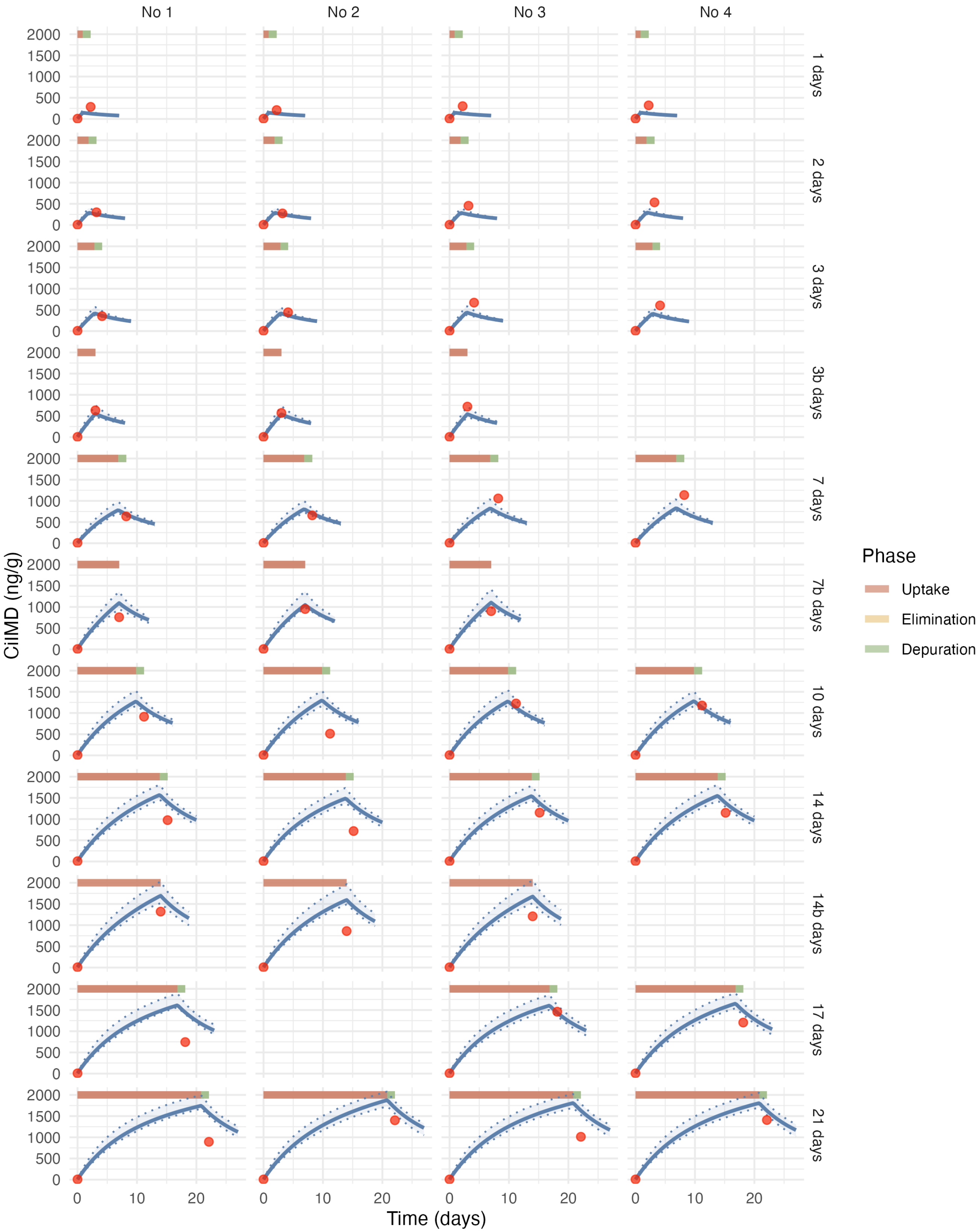

Figure 3.12: Estimated toxicokinetic of the IMD concentration in earthworms exposed to 1000 ng/g of IMD. Red points represent the experimental data, the blue curve represent the estimated toxicokinetic curve and the transparent blue band correspond to the credibility interval of the model.

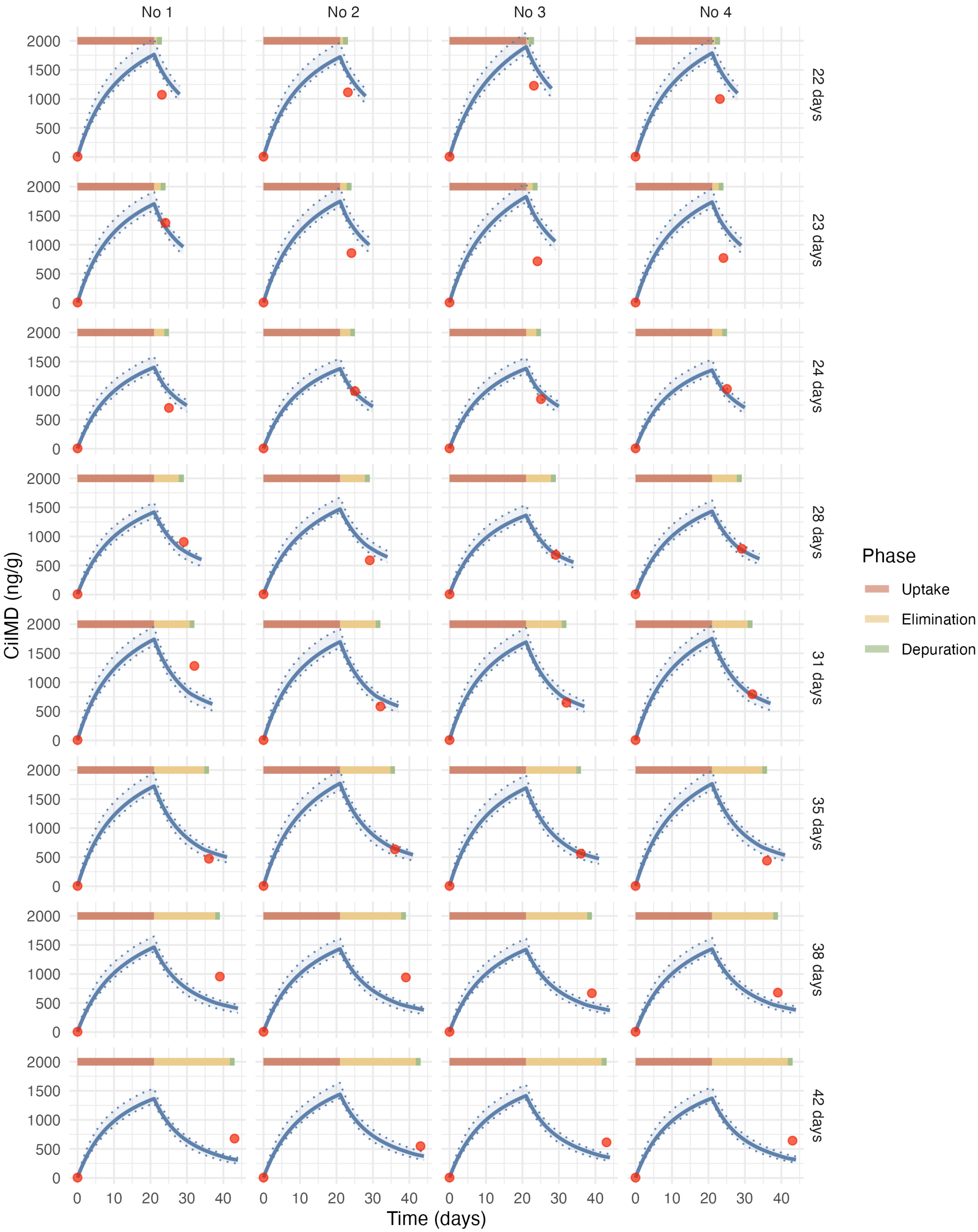

Figure 3.13: Estimated toxicokinetic of the IMD concentration in earthworms exposed to 1000 ng/g of IMD. Red points represent the experimental data, the blue curve represent the estimated toxicokinetic curve and the transparent blue band correspond to the credibility interval of the model.

3.2.5 Model C2P

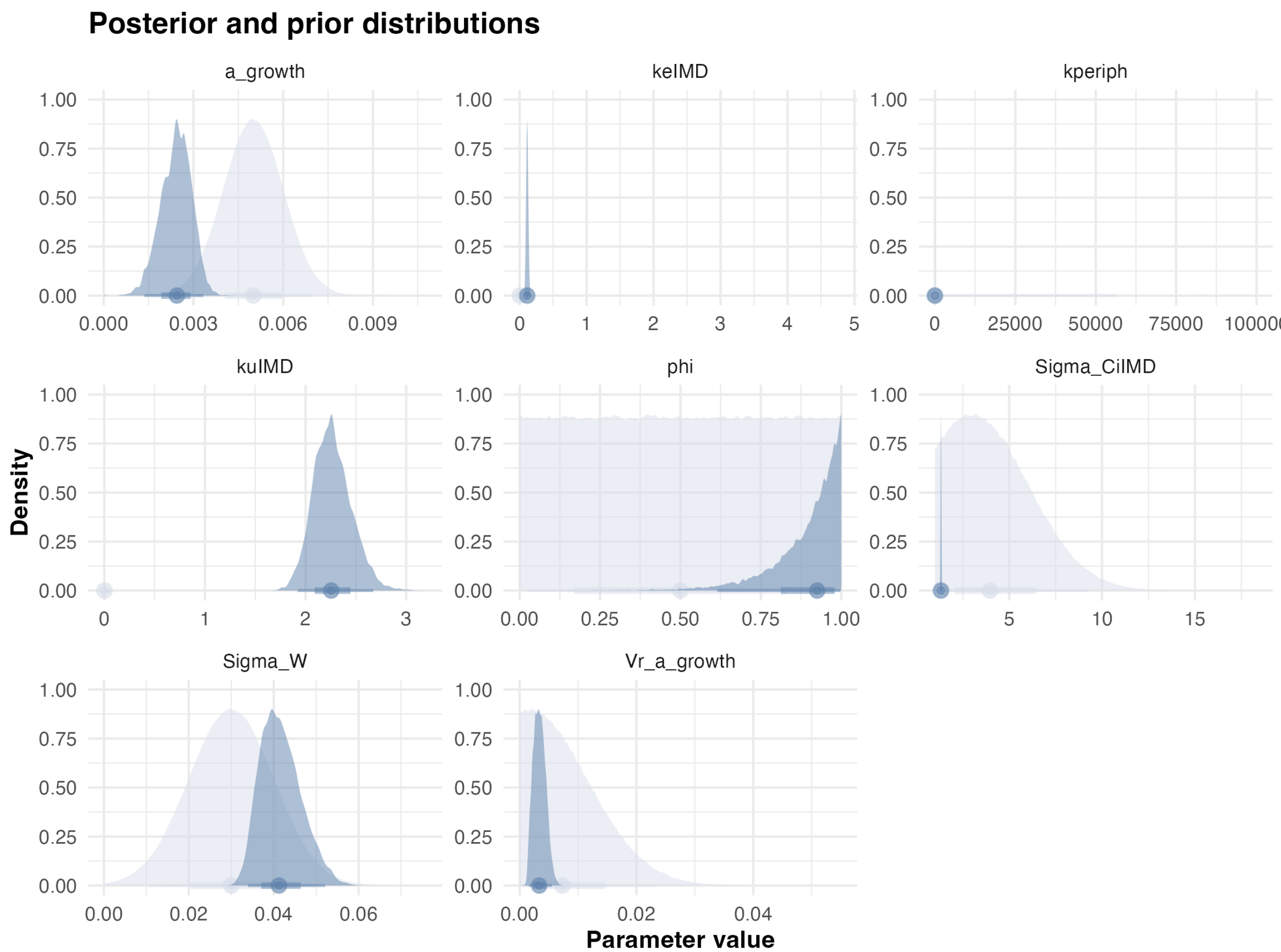

Figure 3.14: Posteriors.

Statistic	k_u	k_e	k_{periph}	ϕ	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	2.2	0.11	0.042	0.97	0.0024	0.002	0.045	1.3	22
Q2.5%	1.9	0.087	0.025	0.62	0.0014	0.0016	0.034	1.3	18
Q97.5%	2.7	0.14	0.065	1	0.0033	0.0056	0.052	1.4	23

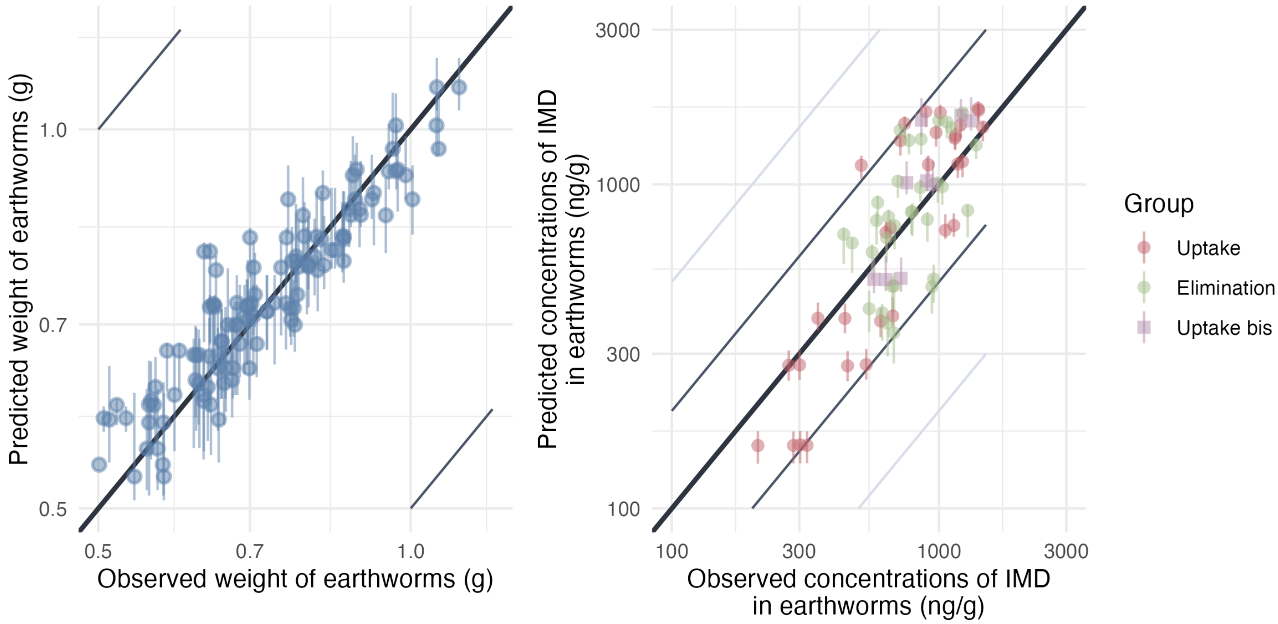

Figure 3.15: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.3 Epoxiconazole

3.3.1 Model C1

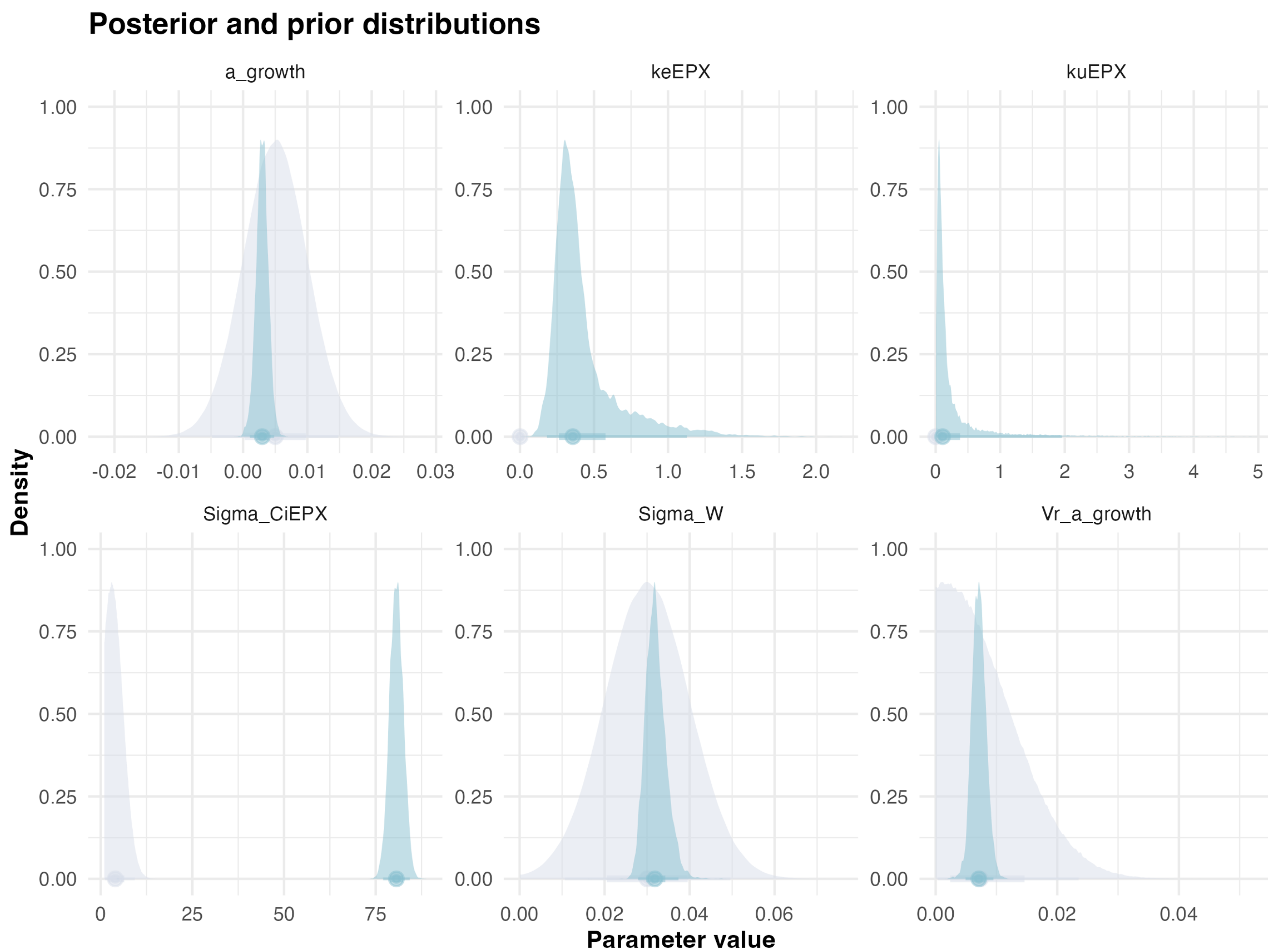

Figure 3.16: Posteriors.

Statistic	k_u	k_e	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	0.028	0.32	0.0039	0.0076	0.028	79	0.097
Q2.5%	0.018	0.18	0.0010	0.0049	0.028	77	0.079
Q97.5%	2.0	1.1	0.0048	0.0095	0.037	84	2

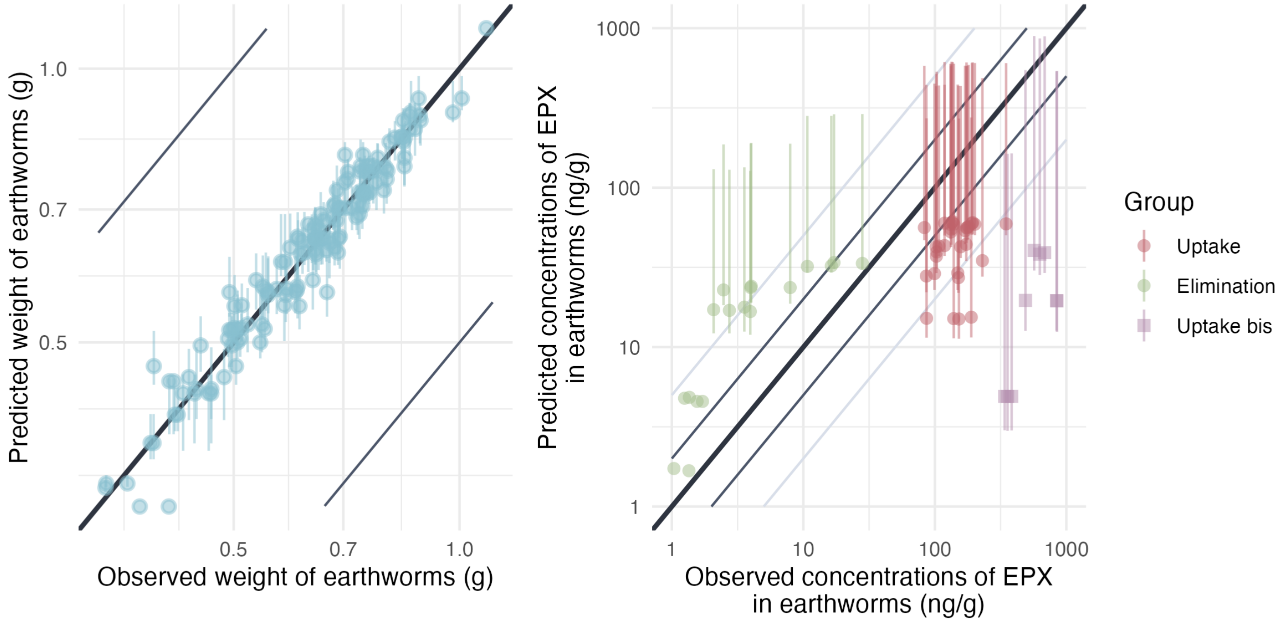

Figure 3.17: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.3.2 Model C1P

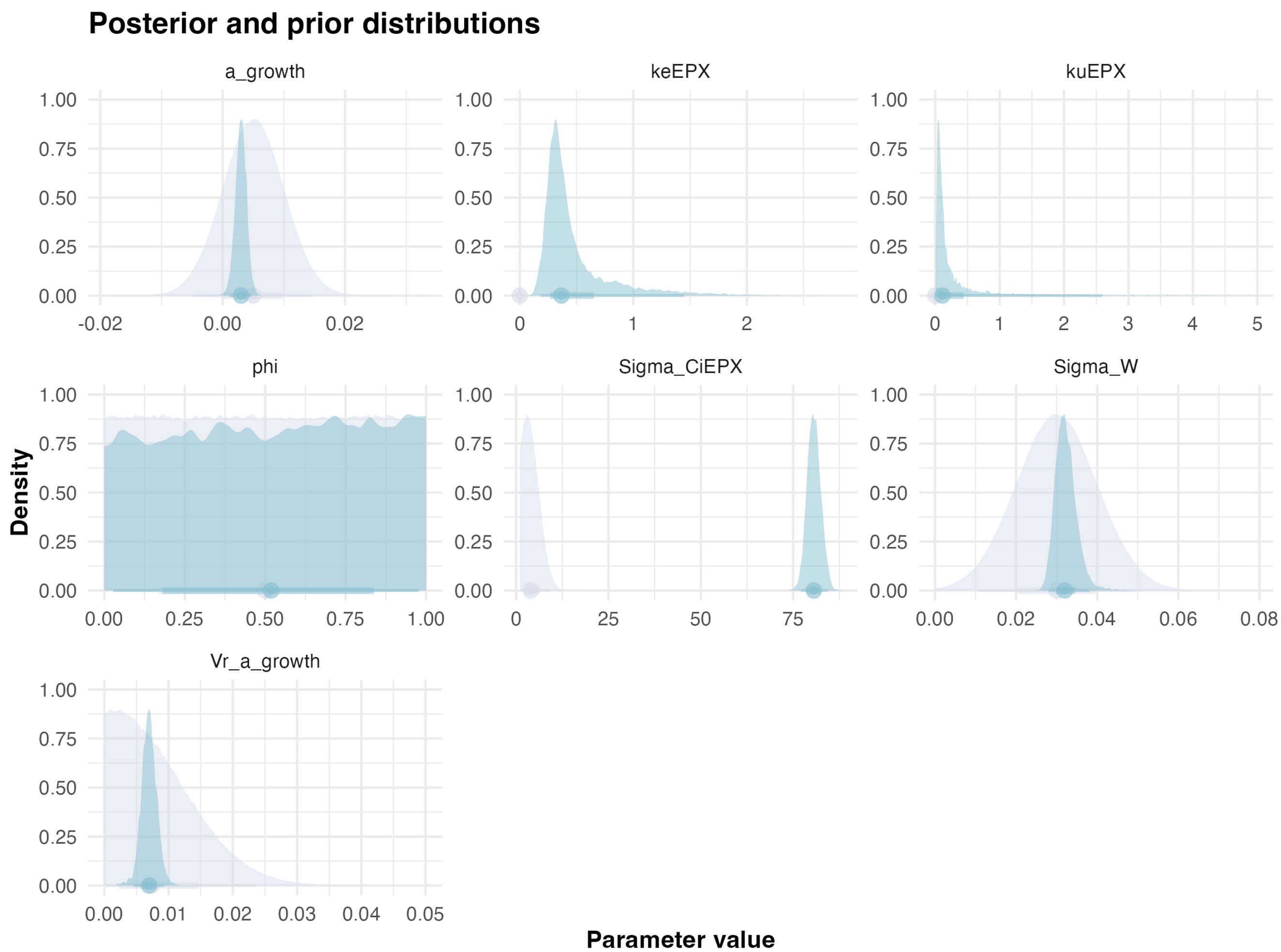

Figure 3.18: Posteriors.

Statistic	k_u	k_e	ϕ	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	0.054	0.21	0.99	0.0022	0.0076	0.030	82	0.18
Q2.5%	0.021	0.18	0.028	0.0011	0.0046	0.028	77	0.0846
Q97.5%	3.5	1.4	0.98	0.0047	0.0093	0.038	84	2.6

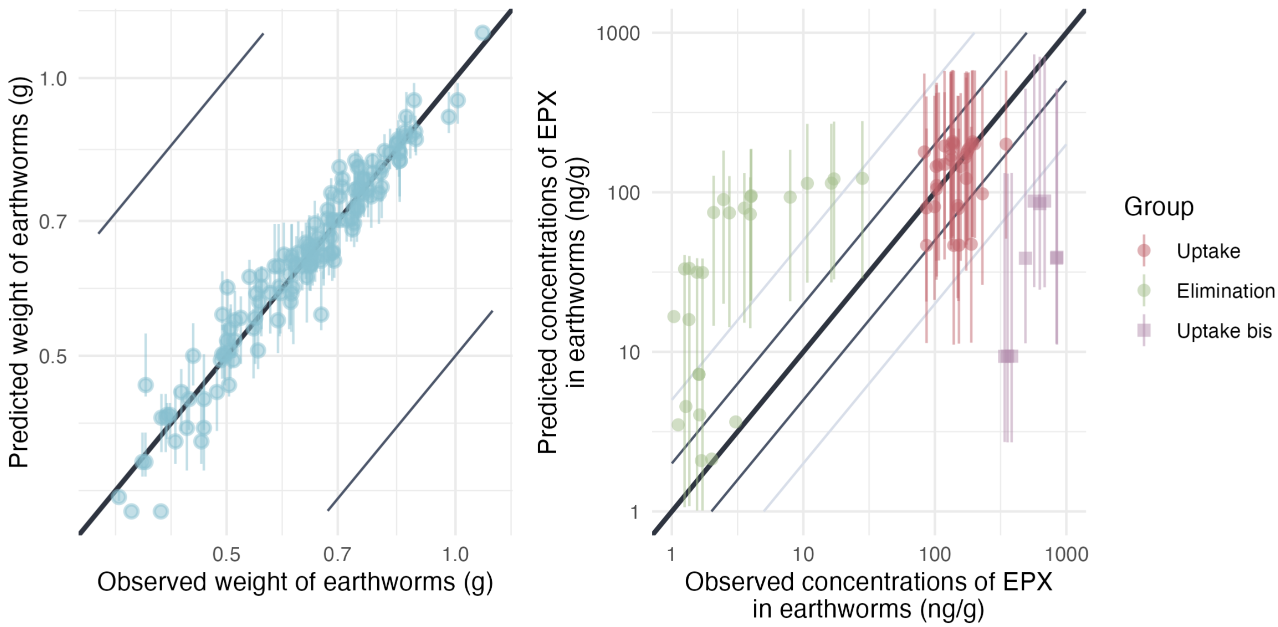

Figure 3.19: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.3.3 Model C2

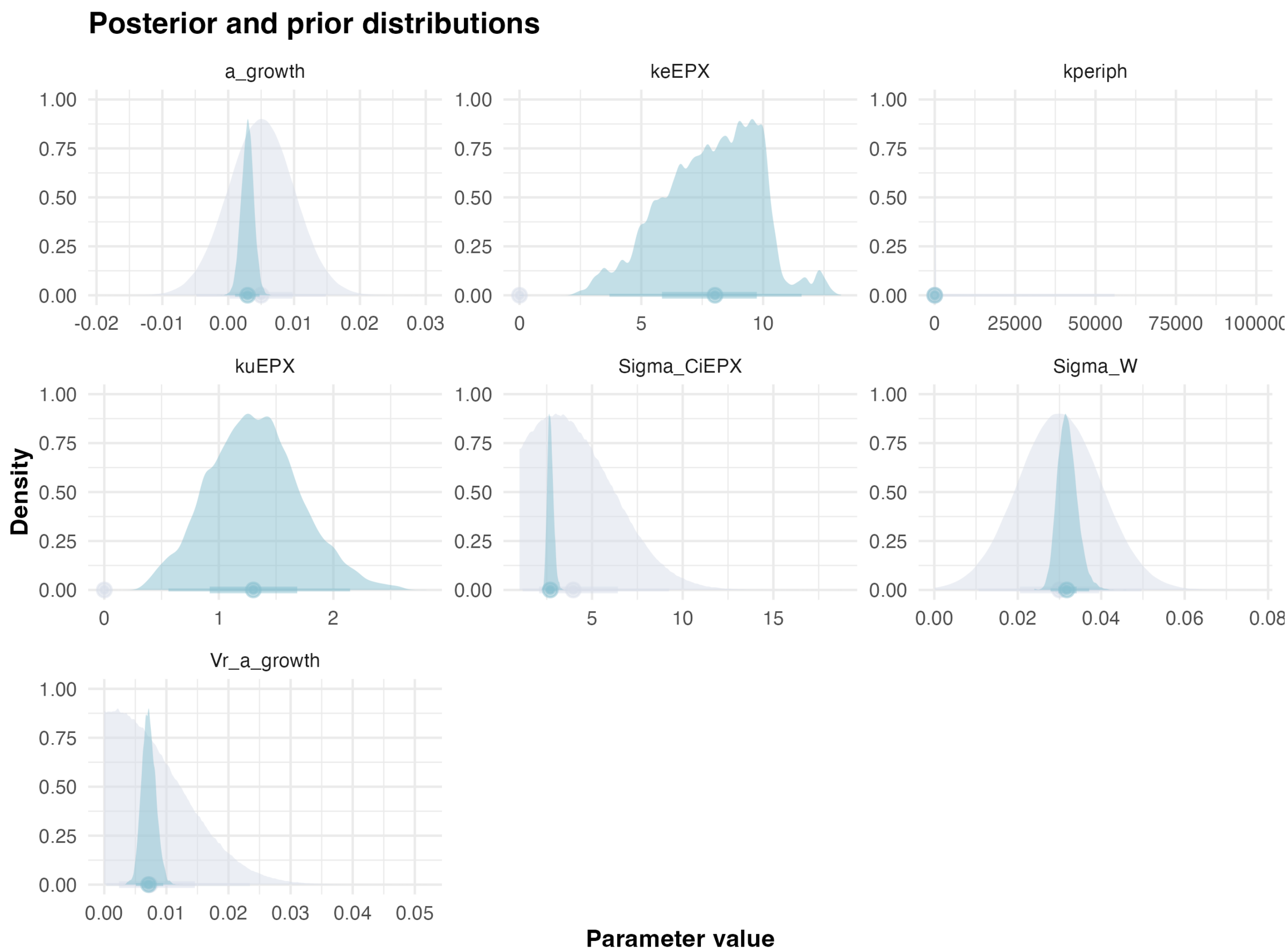

Figure 3.20: Posteriors.

Statistic	k_u	k_e	k_{periph}	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	1.3	8.3	0.28	0.0021	0.0083	0.029	2.5	0.16
Q2.5%	0.56	3.7	0.27	0.0010	0.0051	0.028	2.4	0.12
Q97.5%	2.1	12.0	0.35	0.0048	0.0095	0.037	3.1	0.22

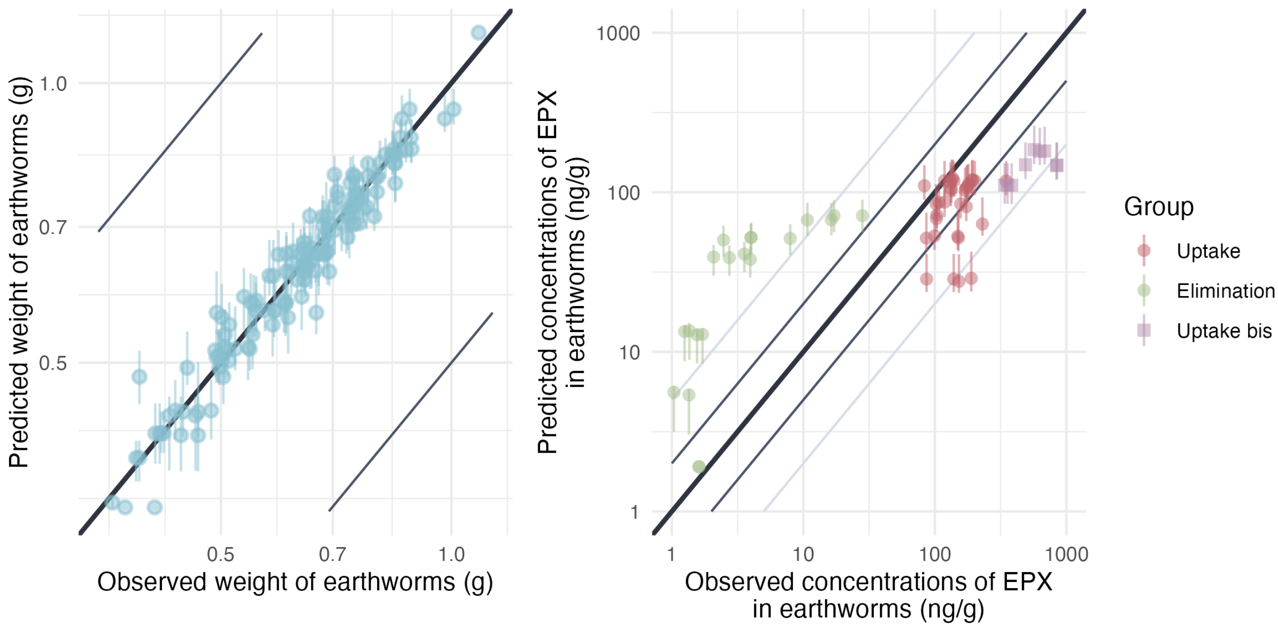

Figure 3.21: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

3.3.4 Model C2P

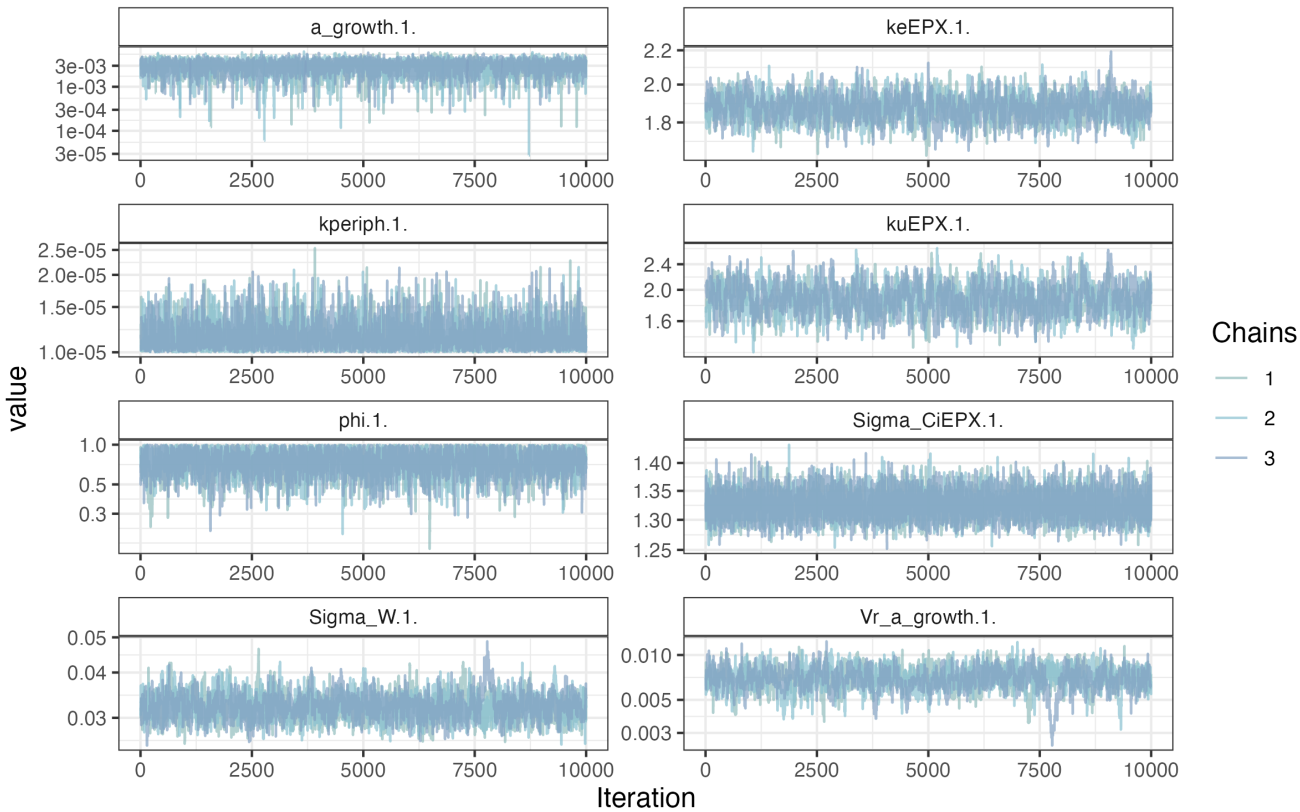

Figure 3.22: MCMC chains of the model.

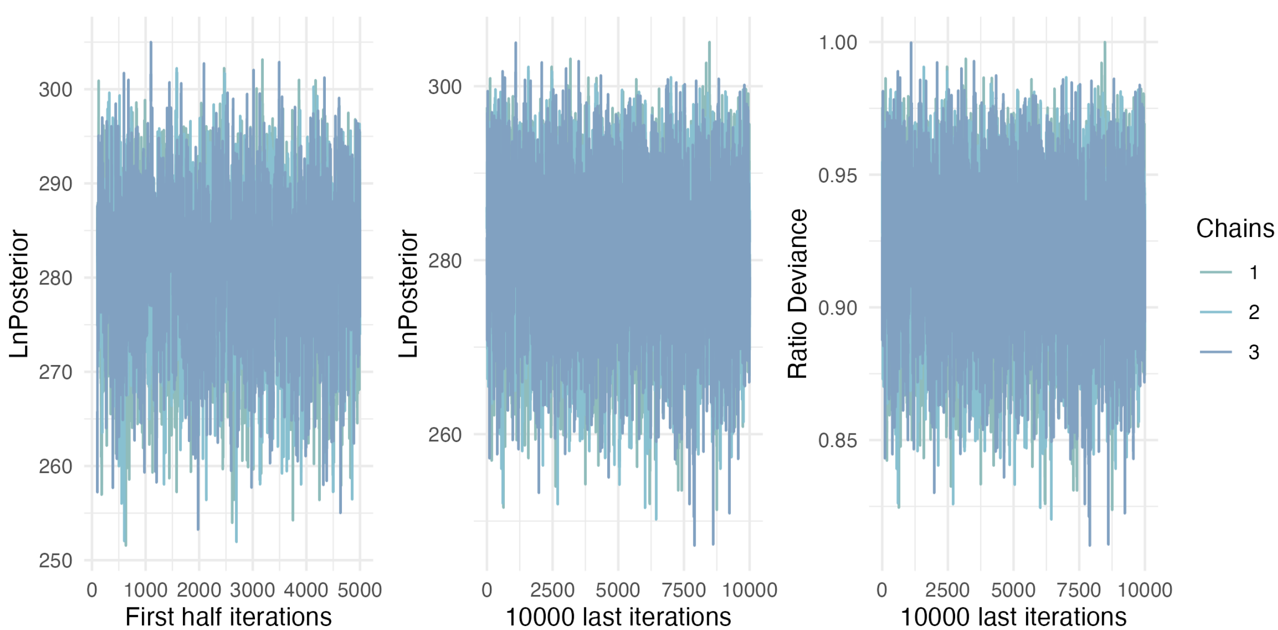

Figure 3.23: Likelihood of chains.

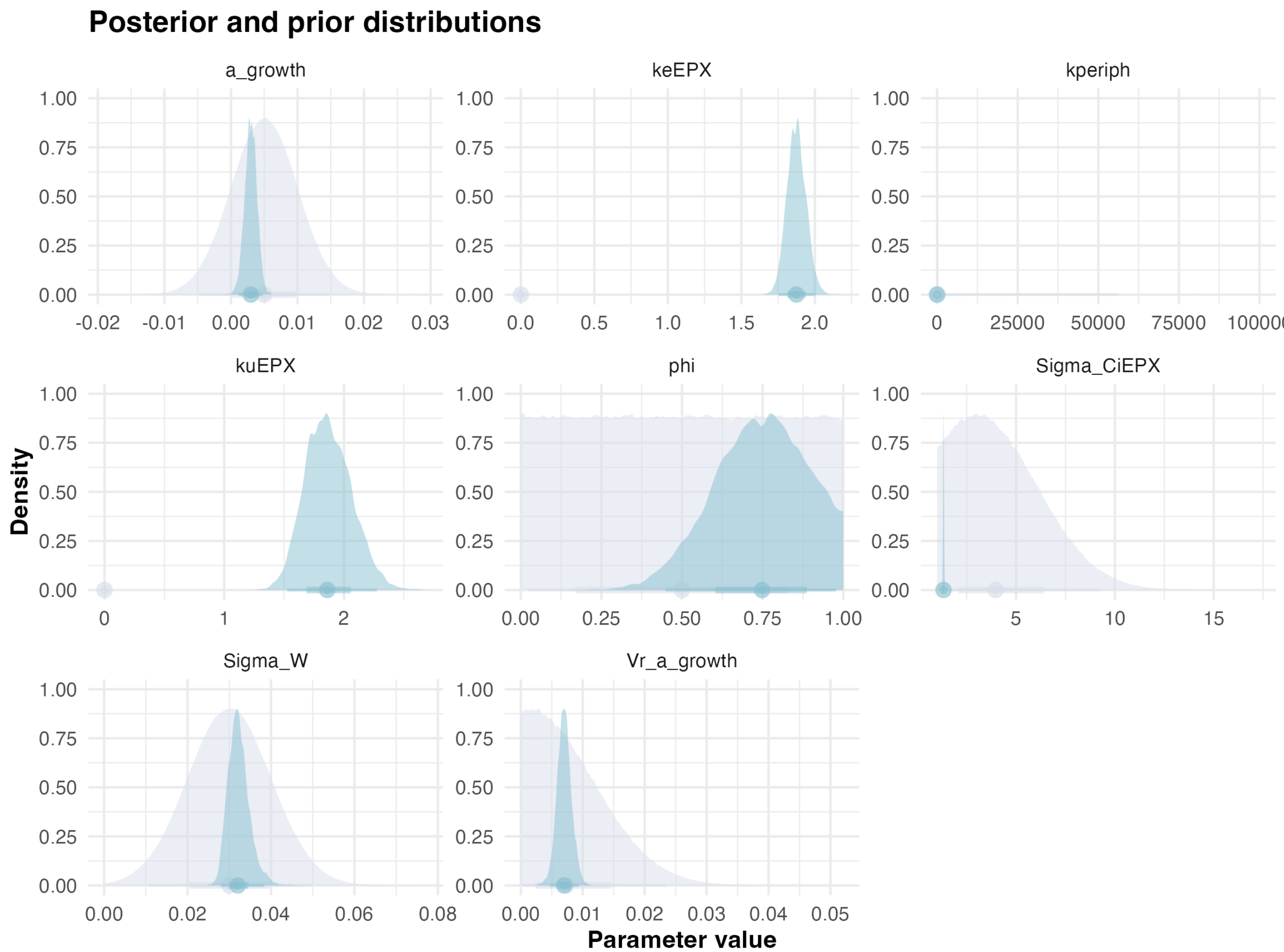

Figure 3.24: Posteriors.

Statistic	k_u	k_e	k_{periph}	ϕ	a_μ	a_σ	W_σ	$C_{i,\sigma}$	BAF
Mode	1.8	1.8	0.000010	0.67	0.0037	0.0076	0.028	1.3	0.87
Q2.5%	1.5	1.8	0.000010	0.45	0.0011	0.0047	0.028	1.3	0.85
Q97.5%	2.3	2.10	0.000015	0.98	0.0047	0.0093	0.038	1.4	1.2

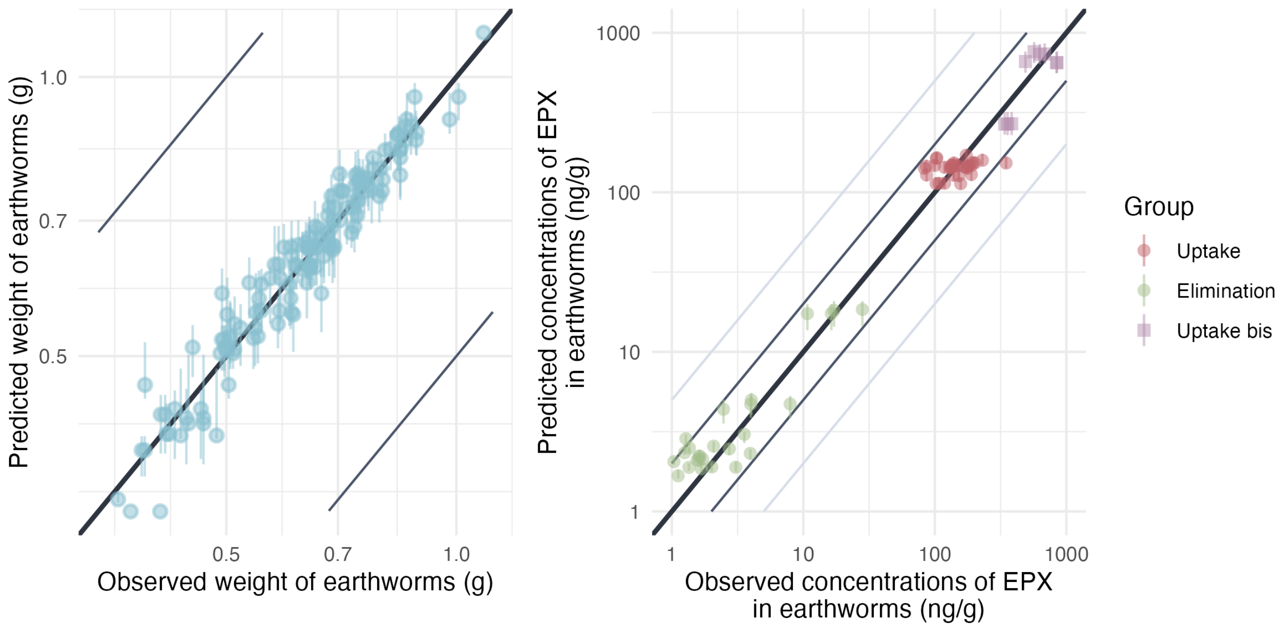

Figure 3.25: Predicted versus observed values of weight without gut content and internal concentrations. Points are colored by exposure scenario: red indicates earthworms exposed only to contaminated soil (uptake phase), green indicates earthworms transferred from contaminated to clean soil (elimination phase) and orange indicates earthworms from the TK BIS experiment. Grey and light-grey lines represent the 2-folds and 5-fold changes, respectively. The bold black line represents the identity line.

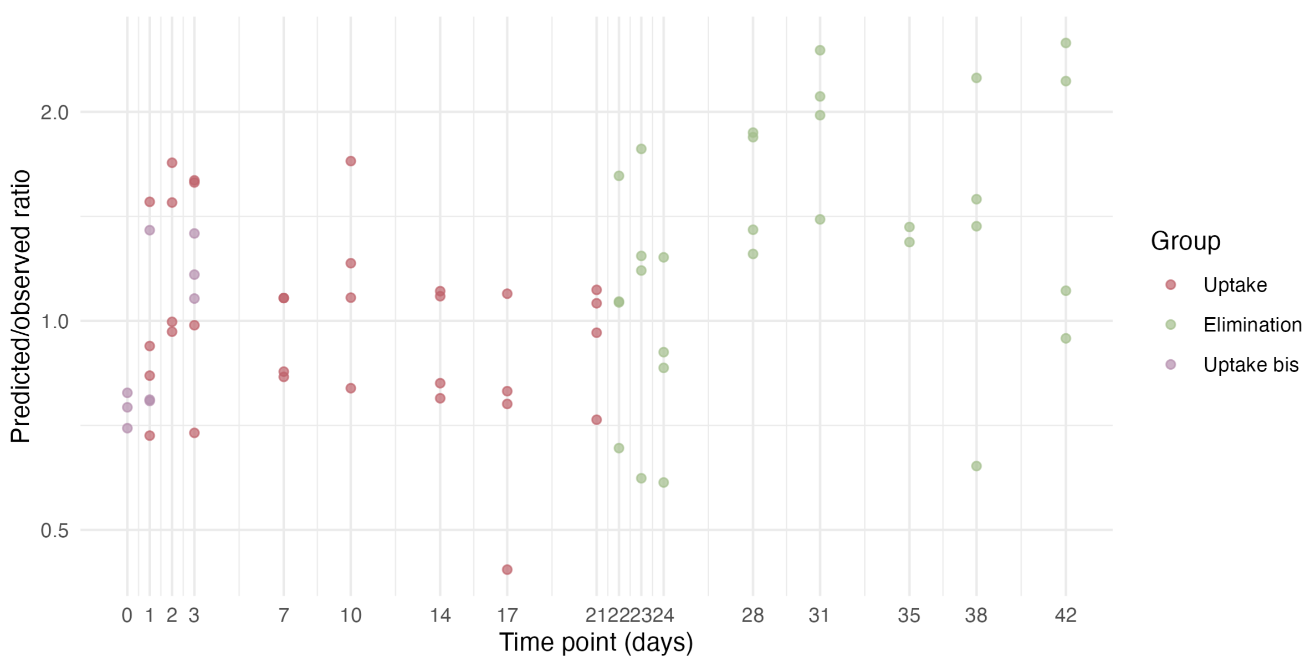

Figure 3.26: Model predicted/observed ratios.

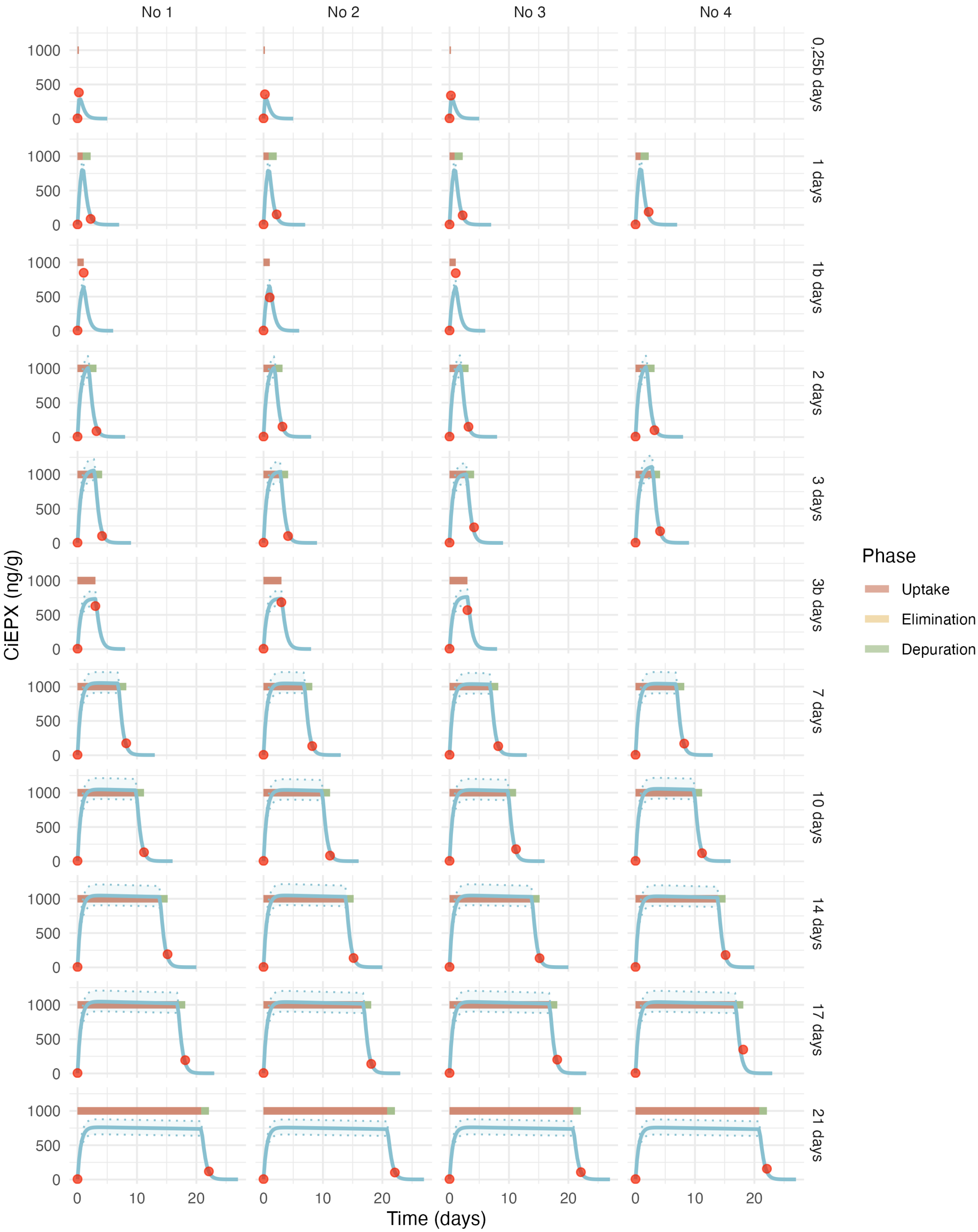

Figure 3.27: Estimated toxicokinetic of the EPX concentration in earthworms exposed to 1000 ng/g of EPX. Red points represent the experimental data, the blue curve represent the estimated toxicokinetic curve and the transparent blue band correspond to the credibility interval of the model.

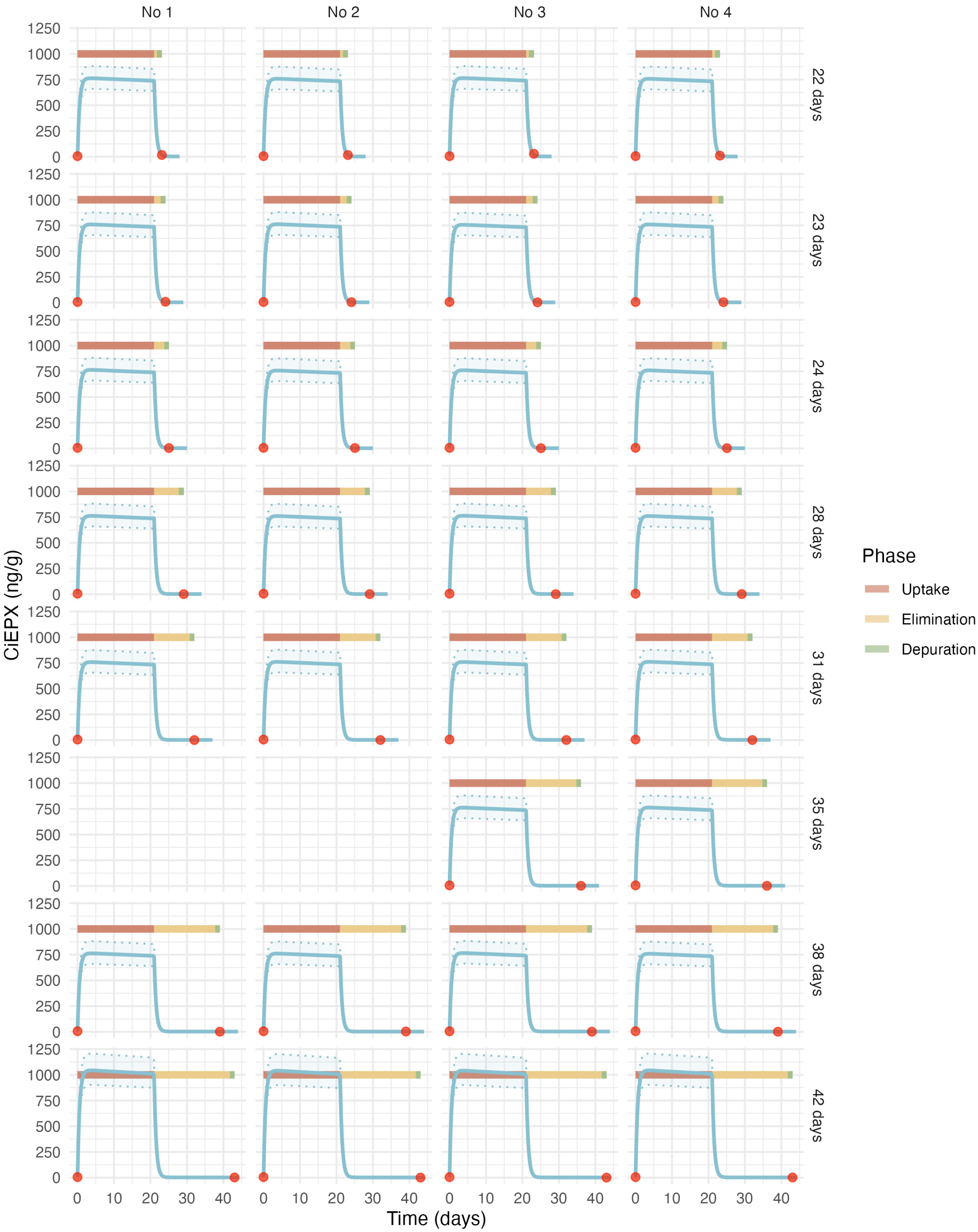

Figure 3.28: Estimated toxicokinetic of the EPX concentration in earthworms exposed to 1000 ng/g of EPX. Red points represent the experimental data, the blue curve represent the estimated toxicokinetic curve and the transparent blue band correspond to the credibility interval of the model.

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